

This is a specimen a_b title^{*,**}

Jos Migchielsen^{1,*}

Elsevier B.V., Radarweg 29, 1043 NX Amsterdam, The Netherlands

CV Radhakrishnan²

Sayahna Foundations, JWRA 34, Jagathy, Trivandrum 695014, India

CV Rajagopal^{1,3}

STM Document Engineering Pvt Ltd., Mepukada, Malayinkil, Trivandrum 695571, India

Abstract

In this work we demonstrate a_b the formation Y₁ of a new type of polariton on the interface between a cuprous oxide slab and a polystyrene micro-sphere placed on the slab. The evanescent field of the resonant whispering gallery mode (WGM) of the micro sphere has a substantial gradient, and therefore effectively couples with the quadrupole 1S excitons in cuprous oxide. This evanescent polariton has a long life-time, which is determined only by its excitonic and WGM component. The polariton lower branch has a well pronounced minimum. This suggests that this excitation is localized and can be utilized for possible BEC. The spatial coherence of the polariton can be improved by assembling the micro-spheres into a linear chain.

Keywords: quadrupole exciton, polariton, WGM, BEC

JEL: 71.35.-y, 71.35.Lk, 71.36.+c

1. Introduction

Although quadrupole excitons (QE) in cuprous oxide crystals are good candidates for BEC due to their narrow line-width and long life-time there are some factors impeding BEC [1, 2]. One of these factors is that due to the small but non negligible coupling to the photon bath, one must consider BEC of the corresponding mixed light-matter states called polaritons [3]. The photon-like part of the polariton has a large group velocity and tends to escape from the crystal. Thus,

*This document is the results of the research project funded by the National Science Foundation.

**The second title footnote which is a longer text matter to fill through the whole text width and overflow into another line in the footnotes area of the first page.

*Corresponding author

Email addresses: J.Migchielsen@elsevier.com (Jos Migchielsen), cvr@sayahna.org (CV Radhakrishnan)

URL: www.stmdocs.in (CV Rajagopal)

¹This is the first author footnote.

²Another author footnote, this is a very long footnote and it should be a really long footnote. But this footnote is not yet sufficiently long enough to make two lines of footnote text.

³Yet another author footnote.

the temporal coherence of the condensate is effectively broken [4, 5]. One proposed solution to this issue is to place the crystal into a planar micro-cavity [6]. But even state-of-the-art planar micro-cavities can hold the light no longer than $10 \mu s$. Besides, formation of the polaritons in the planar cuprous oxide micro-cavity is not effective due to quadrupole origin of the excitons.

Theorem 1. *In this work we demonstrate the formation of a new type of polariton on the interface between a cuprous oxide slab and a polystyrene micro-sphere placed on the slab. The evanescent field of the resonant whispering gallery mode (WGM) of the micro sphere has a substantial gradient, and therefore effectively couples with the quadrupole $1S$ excitons in cuprous oxide. This evanescent polariton has a long life-time, which is determined only by its excitonic and WGM component. The polariton lower branch has a well pronounced minimum. This suggests that this excitation is localized and can be utilized for possible BEC. The spatial coherence of the polariton can be improved by assembling the micro-spheres into a linear chain.*

Therefore in this work we propose to prevent the polariton escaping by trapping it into a whispering gallery mode (WGM)⁴ of a polystyrene micro-sphere (PMS).

We develop a model which demonstrates formation of a strongly *localized* polariton-like quasi-particle. This quasi-particle is formed by the *resonant* interaction between the WGM in PMS and QE in the adjacent layer of cuprous oxide. The QE interacts with the *gradient* of the WGM evanescent field.

There are few experiments concerned with resonant interaction of the WGM and dipole allowed exciton (DE) [7, 8]. But the DE has some disadvantages compared to QE when it comes to interaction with the WGM. First, the evanescent light has small intensity. Therefore it is not effective for the dipole allowed coupling. But it has a large gradient, so it can effectively couple through a quadrupole part. Second, the DE has short life time compared to the QE and therefore is not suitable for BEC. Third, the kinetic energy of the DE is comparable with the interaction energy. Hence the localization is effectively impeded.

2. Evanescent vs. conventional quadrupole light-matter coupling

Assume that a *single* PMS of radius $r_0 \mu m$ is placed at a small⁵ distance $\delta r_0 \ll r_0$ from the cuprous oxide crystal ($\epsilon_{Cu_2O} = 6.5$).

There are several methods to observe WGM-QE interaction. One of them is to mount a prism (or a fiber) on the top of PMS [7]. But any surface nearby perturbs spherical symmetry and therefore reduces the Q factor and the life-time of the WGM. But the positions of the corresponding Mie resonances are not affected. Therefore in this paper we adopt a slightly different picture. Namely the scattering of the bulk polariton in cuprous oxide by the PMS. If one of the Mie resonances is in resonance with the QE one can expect formation of the new type of polariton.

Assume some density of quadrupole $1S$ excitons ([QE]), $\hbar\omega_{1S} = 2.05 eV$, $\lambda_{1S} = 2\pi/\omega_{1S} = 6096 \text{ \AA}$) has been created by an external laser pulse. The corresponding polaritons move in the crystal as the polariton and can be trapped by the PMS due to WGM-QE resonant interaction.

⁴WGM occur at particular resonant wavelengths of light for a given dielectric sphere size. At these wavelengths, the light undergoes total internal reflection at the sphere surface and becomes trapped within the particle for timescales of the order of ns .

⁵comparing to the evanescent field penetration depth

The WGM evanescent field penetration depth into the cuprous oxide adjacent crystal is much larger than the QE radius:

$$\lambda_{1S}/2\pi(\epsilon_{Cu2O} - 1)^{1/2} = 414 \text{ \AA} \gg a_B = 4.6 \text{ \AA}$$

Hence, the light-matter interaction can be considered semi-classically. For the same reason we consider only *bulk* polaritons. However the theory may be expanded to include the surface polaritons also. In the late case the evanescent field of such a surface polariton could play an essential role in the interaction with the WGM, comparable with WGM-QE coupling.

For resonance coupling with a WGM its size parameter should be determined by the resonant wave vector in the cuprous oxide $k_0 = 2.62 \times 10^7 \text{ m}^{-1}$. For example, if one takes a polystyrene (refractive index $\epsilon^2 = 1.59$) sphere of radius $r_0 = 10.7 \text{ \mu m}$ then $k_0 r_0 = 28.78350$. This size parameter corresponds to the 39TE1 resonance [9].

The photon part of the polariton trapped inside the PMS moves as it would move in a micro-cavity of the effective modal volume $V \ll 4\pi r_0^3/3$. Consequently, it can escape through the evanescent field. This evanescent field essentially has a quantum origin and is due to tunneling through the potential caused by dielectric mismatch on the PMS surface. Therefore, we define the *evanescent* polariton (EP) as an evanescent light - QE coherent superposition.

Below we compare the evanescent quadrupole polariton and conventional bulk quadrupole polariton in cuprous oxide. For simplicity let us consider the incident polariton wave vector running along the interface (z direction). The polarization of the polariton is taken along the x direction. Therefore, in the system of coordinates centered at the sphere, the photon part of the incident polariton can be written as [10]:

$$\mathbf{E}_i = \sum_l E_0 i^l \frac{2l+1}{l(l+1)} (\mathbf{M}_{1l} - i\mathbf{N}_{1l}), \quad (1)$$

where $\mathbf{M}_{1l}$ and $\mathbf{N}_{1l}$ are vector spherical harmonics corresponding to TE- and TM- polarized modes of angular momentum l ; the z component of the angular momentum is $|m| = 1$; E_0 is the amplitude of the electric field. The scattered field is given as:

$$\mathbf{E}_s = \sum_l E_0 i^l \frac{2l+1}{l(l+1)} (ia_{1l}\mathbf{N}_{1l} - b_{1l}\mathbf{M}_{1l}), \quad (2)$$

here a_{1l} and b_{1l} are scattering Mie coefficients (See the Appendix). Taking into account that both WGM and QE have narrow line-width, and the energy separation between different WGM is much bigger than the interaction energy we adopt a single mode picture [7]. Keeping only the resonant term the last expression yields:

$$\mathbf{E}_s = -E_0 i^l 0.05 b_{1,39} \mathbf{M}_{1,39}, \quad (3)$$

To calculate the interaction of the plane wave (conventional polariton) (1) and WGM (evanescent polariton) with cuprous oxide one has to change to the cuprous oxide centered system of coordinate (See Fig.2) While in the system of the coordinate, centered at the cuprous oxide, the plane wave is still given by the expression (1), the scattered field has to be changed according to the vector spherical harmonic addition theorem [11]:

$$\mathbf{M}_{1,39} = A_{1,39}^{ml} (r_0 + \delta r) \mathbf{M}_{ml} + B_{1,39}^{ml} (r_0 + \delta r) \mathbf{N}_{ml} \quad (4)$$

Here $A_{1,39}^{ml}$ and $B_{1,39}^{ml}$ are the translational coefficients. Their explicit expression can be found, for instance, in [12, 9] and are explicitly listed in the Appendix.

The bulk (incident) and evanescent polaritons in cuprous oxide are formed through the quadrupole part of the light-matter interaction:

$$H_{int} = \frac{ie}{m\omega_{1S}} \mathbf{E}_{i,s} \cdot \mathbf{p}$$

Here e, m are the electron charge and mass; $\mathbf{p}$ is the electron momentum. For the quadrupole 1S transition in cuprous oxide the energy of interaction can be written as:

$$\sum_{n=0}^{\infty} A_n \int dx \frac{F_n(x)}{A_n + B_n} = B^n C^n \int dx \int dy \frac{G_n(x, y)}{A_n x + B_n y} + \frac{G_n(x, y)}{A_n x + B_n y} \quad (5)$$

Here we introduced the initial state of the system, which transforms as irreducible representation $^1\Gamma_1^+$ of the cubic centered group O_h . The final state is the *ortho*-exciton state which transforms as $^3\Gamma_{5,xz}^+$ in Cartesian system or as $^3\Gamma_{5,1,2}^+$ in the corresponding spherical basis.

Hence, using (1, 3, 4, 5), one can deduce that the coupling of the spherical harmonic compared to the plane wave ($\hbar g_{1,2} = 124 \mu eV$) is resonantly enhanced:

$$\frac{g_{1,39}}{g_{1,2}} = -i0.06b_{1,39}(kr_0)A_{1,39}^{1,2}(r_0 + \delta r) \quad (6)$$

Here we utilized the fact that $B_{1,39}^{1,2} \ll A_{1,39}^{1,2}$. While the resonant enhancement is provided by the $b_{1,39}$ Mie coefficient here, the translational coefficient reduces the effect. That is why if one tries to couple the evanescent light to the dipole transition the effect is much weaker as $A_{1,39}^{0,1} \ll A_{1,39}^{1,2}$. The resulting exciton - evanescent light coupling is shown in Fig.1 Both dipole and quadrupole



Figure 1: The evanescent light - 1S quadrupole coupling ($g_{1,l}$) scaled to the bulk exciton-photon coupling ($g_{1,2}$). The size parameter kr_0 is denoted as x and the PMS is placed directly on the cuprous oxide sample ($\delta r = 0$, See also Fig.2).

coupling rate in the actual combined semiconductor-microsphere system is smaller than that in case of conventional polariton. This is attributed to the fact that the coupling occurs in a small region of the evanescent tail penetrating into cuprous oxide, although the coupling grows with mode number l , because the gradient of the evanescent field increases. Note that QE realizes

strong coupling regime $g_{1,39} > \gamma$ while DE demonstrates *weak* regime only [7]. The property of the scalable coupling factor can be utilized in practical applications such as non-linear optics and is the subject of our future work.

3. Results and discussion

In this section let us utilize the above calculated WGM-QE interaction to obtain the evanescent polariton (EP) dispersion in the framework of the coupled oscillator model that has been widely used for describing coupled atom-photon or exciton-photon modes in microcavity systems [13]. Near the resonance between WGM and the quadrupole exciton $\omega_{1l} \approx \omega_{1S}$ the EP branches are given by the eigenvalues of the following Hamiltonian:

$$H/\hbar = \omega_{1l} a_x^\dagger a_x + \omega_{1S} b_x^\dagger b_x + g_{1l}(x) (a_k^\dagger b_x + a_x b_x^\dagger), \quad (7)$$

here a_x , b_x are annihilation operators for light and the exciton, respectively. We also neglected kinetic energy of the QE due to smallness of the resonant wave vector and big mass of the QE. Therefore, considering that both the exciton and WGM of a single sphere are localized, the dispersion is reduced to:

$$\omega = \omega_{1S} \pm g_{1l}/\hbar \quad (8)$$

The above expression shows the formation of the doublet at resonance (both states are exactly half-QE, half-WGM). Recall that for DE-WGM weak coupling only WGM pattern shifted by the coupling has been observed [7].

The excitons are trapped in the minimum of the lower branch thus populating the strongly localized states. Physically this means that the resonant coupling with localized WGM does not let QE escape by means of its kinetic energy.

The dispersion above is similar to the quadrupole-dipole hybrid in the organic-inorganic hetero-structures [2]. In the later case, the excited organic molecules create an evanescent field penetrating into the cuprous oxide.

Now let us consider possible application of the evanescent polariton to BEC. The problem of the conventional polariton escaping from the crystal mentioned in the introduction no longer exists for the localized states of the evanescent polariton.

The linewidth of the mixed state is expected to be in the first approximation the sum of the exciton and photon linewidth weighted by the exciton and photon component of the polariton. At resonance, the linewidth of the two eigenstates is simply given by $\gamma = (\gamma_{1S} + \gamma_{m,l})/2$. Where γ_{1S} and $\gamma_{m,l}$ are the QE and WGM linewidth. Even having taken into account decrease of the Q factor due to PMS contact with the cuprous oxide sample $\gamma_{1S} \gg \gamma_{1,39}$ for the ortho-exciton. Hence, γ is defined by the QE linewidth.

For the Cu_2O para-exciton the linewidth of the polariton is given by the WGM linewidth. The para-excitons can acquire some oscillator strength provided the PMS exerts a local stress upon the cuprous oxide sample. The applied stress changes the crystal symmetry, so that usually optically inactive para-excitons may couple to the WGM. Note that due to spin-orbit interaction the para-excitons are $12 \mu eV$ below the ortho-exciton. Therefore for given radius of the PMS the resonant interaction occurs with different WGM for ortho- and para- cases.

The evanescent polariton provided by a single sphere gives the time coherence necessary for the observable BEC of the quadrupole exciton. But the spatial coherence is limited to a small region near the sphere. To improve the spatial coherence one has to sacrifice the temporal



Figure 2: Schematic of formation of the evanescent polariton on linear chain of PMS. The actual dispersion is determined by the ratio of two coupling parameters such as exciton-WGM coupling and WGM-WGM coupling between the microspheres.

coherence slightly by delocalizing the corresponding WGM. It can be done by using an array of spheres aligned along the z direction and separated by the distance δr_0 (See Fig.2).

Recent experimental [14] and theoretical [15] studies have shown that the WGM can travel along the chain as "heavy photons". Therefore the WGM acquires the spatial dispersion, and the evanescent quadrupole polariton has the form (See Fig.3):

$$2\omega = \omega_{1l,k} + \omega_{1s} \pm \sqrt{(\omega_{1l,k} - \omega_{1s})^2 + 4|g_{1l}/\hbar|^2}$$

$$\omega_{1l,k} = \omega_{1s} + 2\left(g_{1l}^I/\hbar\right)\cos(x - x_{1l} + \pi/2) \quad (9)$$

Here $g_{1l}^I = \omega_{1s} b_{1l} A_{1l}^I(\delta r_1)$ is the nearest-neighbor inter-sphere coupling parameter.



Figure 3: Dispersion of the evanescent polariton 39TE1. The dashed line (1) corresponds to the dispersion of the chain of spheres touching each other ($\delta r_0 = 0$). The thin solid line (3) stands for upper and lower branches of a single sphere dispersion ($\delta r_0 \gg \delta r = 0$). The thick solid curve (2) is the case of linear chain of the spheres in contact with the cuprous oxide ($\delta r_0 = \delta r = 0$).

When the coupling between spheres dominates ($\delta r \gg \delta r_0$) the minimum of the lower polariton branch disappears. Consequently, for possible BEC of the evanescent polariton one has

to keep the desired balance between spatial and temporal coherence by adjusting experimental parameters δr and δr_0 .

Both, the energy of the $1S$ quadrupole exciton and the WGM depend on the temperature. Therefore one can use a standard temperature scan to reveal the evanescent polariton dispersion [16].

In summary, we note that there is some similarity between BEC of alkali atoms trapped by the laser field [17] and the long living QE localized by the resonant WGM.

The theory developed above is applicable also for void cavities, spherical impurities and metallic droplets in bulk cuprous oxide crystal.

4. Appendix

In the appendix we list explicit expression for the Mie scattering coefficient:

$$a_{ml} = \frac{n^2 j_{ml}(nx) [x j_{ml}(x)]' - j_{ml}(x) [nx j_{ml}(nx)]'}{n^2 j_{ml}(nx) [x h_{ml}^{(1)}(x)]' - h_{ml}^{(1)}(x) [nx j_{ml}(nx)]'}$$

$$b_{ml} = \frac{j_{ml}(nx) [x j_{ml}(x)]' - n^2 j_{ml}(x) [nx j_{ml}(nx)]'}{j_{ml}(nx) [x h_{ml}^{(1)}(x)]' - n^2 h_{ml}^{(1)}(x) [nx j_{ml}(nx)]'}$$

Here $n = \epsilon^2$ is the refractive index of the spheres; $x = kr_0$ is the size parameter; j_{ml} , h_{ml} are the spherical Bessel and Hankel of the first kind functions respectively.

In the case of $l \gg 1$ the calculation of the translational coefficients can be significantly simplified with the help of the so-called maximum term approximation [9].

$$A_l' \cong -2l(-1)^{l+1} \sqrt{\frac{l+l'}{\pi(l'+1)(l-1)}} \times$$

$$\frac{l'(l')'}{(l'+1)^{l'+1} (l-1)^{l-1}} h_{l+l'}^{(1)}(\eta x)$$

$$B_l' \cong i \frac{x|i-j|}{ll'} A_l'$$

Here η defined as $\eta = |r_0 + \delta r|/r_0 \geq 1$ is a dimensionless distance between the centers of the spheres.

References

- [1] G. Kavoulakis and G. Baym, Phys. Rev. B **53**, 7227 (1996).
- [2] O. Roslyak and J. Birman, arXiv:cond-mat/0703650, PRB to be published (2007).
- [3] D. Frohlich, G. Dasbach, G. B. Hogerthal, M. Bayer, R. Kliebera, D. Sutura, and H. Stolz, Solid State Communications **134**, 139 (2005).
- [4] C. Ell, A. L. Ivanov, and H. Haug, Phys. Rev. B **57**, 9663 (1998).
- [5] D. Snoke, Science **298**, 1368 (2002).
- [6] J. Kasprzak, M. Richard, S. Kundermann, A. Baas, P. Jeambrun, J. Keeling, F. Marchetti, M. Szymanska, R. Andre, J. Staehli, et al., Nature **443**, 409 (2006).
- [7] H. W. Xudong Fan, Scott Lacey, Optics Letters **24**, 771 (1999).
- [8] X. Fan, S. Lacey, and H. Wang, Opt. Lett **24**, 771 (1999).
- [9] H. Miyazaki and Y. Jimba, Phys. Rev. B **62**, 7976 (2000).

- [10] C. Bohren and D. Huffman, *Absorption and scattering of light by small particles* (Wiley New York, 1983).
- [11] S. Stein, Q. appl. Math **19**, 15 (1961).
- [12] K. Fuller, Appl. Opt **30**, 4716 (1991).
- [13] H. J. Carmichael, Phys. Rev. A **33**, 3262 (1986).
- [14] Y. Hara, T. Mukaiyama, K. Takeda, and M. Kuwata-Gonokami, Physical Review Letters **94**, 203905 (2005).
- [15] L. Deych and A. Roslyak, Physical Review E **73**, 36606 (2006).
- [16] E. Peter, P. Senellart, D. Martrou, A. Lemaître, J. Hours, J. Gérard, and J. Bloch, Physical Review Letters **95**, 67401 (2005).
- [17] A. J. Leggett, Rev. Mod. Phys. **73**, 307 (2001).

This is a specimen a_b title^{*,**}

Jos Migchielsen^{a,1,*}, CV Radhakrishnan^{b,2}, CV Rajagopal^{c,1,3}

^aElsevier B.V., Radarweg 29, 1043 NX Amsterdam, The Netherlands

^bSayahna Foundations, JWRA 34, Jagathy, Trivandrum 695014, India

^cSTM Document Engineering Pvt Ltd., Mepukada, Malayinkil, Trivandrum 695571, India

Abstract

In this work we demonstrate a_b the formation Y₁ of a new type of polariton on the interface between a cuprous oxide slab and a polystyrene micro-sphere placed on the slab. The evanescent field of the resonant whispering gallery mode (WGM) of the micro sphere has a substantial gradient, and therefore effectively couples with the quadrupole 1S excitons in cuprous oxide. This evanescent polariton has a long life-time, which is determined only by its excitonic and WGM component. The polariton lower branch has a well pronounced minimum. This suggests that this excitation is localized and can be utilized for possible BEC. The spatial coherence of the polariton can be improved by assembling the micro-spheres into a linear chain.

Keywords: quadrupole exciton, polariton, WGM, BEC

JEL: 71.35.-y, 71.35.Lk, 71.36.+c

1. Introduction

Although quadrupole excitons (QE) in cuprous oxide crystals are good candidates for BEC due to their narrow line-width and long life-time there are some factors impeding BEC [1, 2]. One of these factors is that due to the small but non negligible coupling to the photon bath, one must consider BEC of the corresponding mixed light-matter states called polaritons [3]. The photon-like part of the polariton has a large group velocity and tends to escape from the crystal. Thus, the temporal coherence of the condensate is effectively broken [4, 5]. One proposed solution to this issue is to place the crystal into a planar micro-cavity [6]. But even state-of-the-art planar micro-cavities can hold the light no longer than 10 μ s. Besides, formation of the polaritons in the planar cuprous oxide micro-cavity is not effective due to quadrupole origin of the excitons.

*This document is the results of the research project funded by the National Science Foundation.

**The second title footnote which is a longer text matter to fill through the whole text width and overflow into another line in the footnotes area of the first page.

*Corresponding author

Email addresses: J.Migchielsen@elsevier.com (Jos Migchielsen), cvr@sayahna.org (CV Radhakrishnan)

URL: www.stmdocs.in (CV Rajagopal)

¹This is the first author footnote.

²Another author footnote, this is a very long footnote and it should be a really long footnote. But this footnote is not yet sufficiently long enough to make two lines of footnote text.

³Yet another author footnote.

Theorem 1. *In this work we demonstrate the formation of a new type of polariton on the interface between a cuprous oxide slab and a polystyrene micro-sphere placed on the slab. The evanescent field of the resonant whispering gallery mode (WGM) of the micro sphere has a substantial gradient, and therefore effectively couples with the quadrupole 1S excitons in cuprous oxide. This evanescent polariton has a long life-time, which is determined only by its excitonic and WGM component. The polariton lower branch has a well pronounced minimum. This suggests that this excitation is localized and can be utilized for possible BEC. The spatial coherence of the polariton can be improved by assembling the micro-spheres into a linear chain.*

Therefore in this work we propose to prevent the polariton escaping by trapping it into a whispering gallery mode (WGM)⁴ of a polystyrene micro-sphere (PMS).

We develop a model which demonstrates formation of a strongly *localized* polariton-like quasi-particle. This quasi-particle is formed by the *resonant* interaction between the WGM in PMS and QE in the adjacent layer of cuprous oxide. The QE interacts with the *gradient* of the WGM evanescent field.

There are few experiments concerned with resonant interaction of the WGM and dipole allowed exciton (DE) [7, 8]. But the DE has some disadvantages compared to QE when it comes to interaction with the WGM. First, the evanescent light has small intensity. Therefore it is not effective for the dipole allowed coupling. But it has a large gradient, so it can effectively couple through a quadrupole part. Second, the DE has short life time compared to the QE and therefore is not suitable for BEC. Third, the kinetic energy of the DE is comparable with the interaction energy. Hence the localization is effectively impeded.

2. Evanescent vs. conventional quadrupole light-matter coupling

Assume that a *single* PMS of radius r_0 μm is placed at a small⁵ distance $\delta r_0 \ll r_0$ from the cuprous oxide crystal ($\epsilon_{\text{Cu}_2\text{O}} = 6.5$).

There are several methods to observe WGM-QE interaction. One of them is to mount a prism (or a fiber) on the top of PMS [7]. But any surface nearby perturbs spherical symmetry and therefore reduces the Q factor and the life-time of the WGM. But the positions of the corresponding Mie resonances are not affected. Therefore in this paper we adopt a slightly different picture. Namely the scattering of the bulk polariton in cuprous oxide by the PMS. If one of the Mie resonances is in resonance with the QE one can expect formation of the new type of polariton.

Assume some density of quadrupole 1S excitons ([QE]), $\hbar\omega_{1S} = 2.05 \text{ eV}$, $\lambda_{1S} = 2\pi/\omega_{1S} = 6096 \text{ \AA}$ has been created by an external laser pulse. The corresponding polaritons move in the crystal as the polariton and can be trapped by the PMS due to WGM-QE resonant interaction.

The WGM evanescent field penetration depth into the cuprous oxide adjacent crystal is much larger than the QE radius:

$$\lambda_{1S}/2\pi(\epsilon_{\text{Cu}_2\text{O}} - 1)^{1/2} = 414 \text{ \AA} \gg a_B = 4.6 \text{ \AA}$$

⁴WGM occur at particular resonant wavelengths of light for a given dielectric sphere size. At these wavelengths, the light undergoes total internal reflection at the sphere surface and becomes trapped within the particle for timescales of the order of ns .

⁵comparing to the evanescent field penetration depth

Hence, the light-matter interaction can be considered semi-classically. For the same reason we consider only *bulk* polaritons. However the theory may be expanded to include the surface polaritons also. In the late case the evanescent field of such a surface polariton could play an essential role in the interaction with the WGM, comparable with WGM-QE coupling.

For resonance coupling with a WGM its size parameter should be determined by the resonant wave vector in the cuprous oxide $k_0 = 2.62 \times 10^7 \text{ m}^{-1}$. For example, if one takes a polystyrene (refractive index $\epsilon^2 = 1.59$) sphere of radius $r_0 = 10.7 \text{ } \mu\text{m}$ then $k_0 r_0 = 28.78350$. This size parameter corresponds to the 39TE1 resonance [9].

The photon part of the polariton trapped inside the PMS moves as it would move in a micro-cavity of the effective modal volume $V \ll 4\pi r_0^3/3$. Consequently, it can escape through the evanescent field. This evanescent field essentially has a quantum origin and is due to tunneling through the potential caused by dielectric mismatch on the PMS surface. Therefore, we define the *evanescent* polariton (EP) as an evanescent light - QE coherent superposition.

Below we compare the evanescent quadrupole polariton and conventional bulk quadrupole polariton in cuprous oxide. For simplicity let us consider the incident polariton wave vector running along the interface (z direction). The polarization of the polariton is taken along the x direction. Therefore, in the system of coordinates centered at the sphere, the photon part of the incident polariton can be written as [10]:

$$\mathbf{E}_i = \sum_l E_0 i^l \frac{2l+1}{l(l+1)} (\mathbf{M}_{1l} - i\mathbf{N}_{1l}), \quad (1)$$

where $\mathbf{M}_{1l}$ and $\mathbf{N}_{1l}$ are vector spherical harmonics corresponding to TE- and TM- polarized modes of angular momentum l ; the z component of the angular momentum is $|m| = 1$; E_0 is the amplitude of the electric field. The scattered field is given as:

$$\mathbf{E}_s = \sum_l E_0 i^l \frac{2l+1}{l(l+1)} (ia_{1l}\mathbf{N}_{1l} - b_{1l}\mathbf{M}_{1l}), \quad (2)$$

here a_{1l} and b_{1l} are scattering Mie coefficients (See the Appendix). Taking into account that both WGM and QE have narrow line-width, and the energy separation between different WGM is much bigger then the interaction energy we adopt a single mode picture [7]. Keeping only the resonant term the last expression yields:

$$\mathbf{E}_s = -E_0 i^l 0.05 b_{1,39} \mathbf{M}_{1,39}, \quad (3)$$

To calculate the interaction of the plane wave (conventional polariton) (1) and WGM (evanescent polariton) with cuprous oxide one has to change to the cuprous oxide centered system of coordinate (See Fig.2) While in the system of the coordinate, centered at the cuprous oxide, the plane wave is still given by the expression (1), the scattered field has to be changed according to the vector spherical harmonic addition theorem [11]:

$$\mathbf{M}_{1,39} = A_{1,39}^{ml} (r_0 + \delta r) \mathbf{M}_{ml} + B_{1,39}^{ml} (r_0 + \delta r) \mathbf{N}_{ml} \quad (4)$$

Here $A_{1,39}^{ml}$ and $B_{1,39}^{ml}$ are the translational coefficients. Their explicit expression can be found, for instance, in [12, 9] and are explicitly listed in the Appendix.

The bulk (incident) and evanescent polaritons in cuprous oxide are formed through the quadrupole part of the light-matter interaction:

$$H_{int} = \frac{ie}{m\omega_{1S}} \mathbf{E}_{i,s} \cdot \mathbf{p}$$

Here e, m are the electron charge and mass; $\mathbf{p}$ is the electron momentum. For the quadrupole $1S$ transition in cuprous oxide the energy of interaction can be written as:

$$\sum_{n=0}^{\infty} A_n \int dx \frac{F_n(x)}{A_n + B_n} = B^n C^n \int dx \int dy \frac{G_n(x, y)}{A_n x + B_n y} + \frac{G_n(x, y)}{A_n x + B_n y} \quad (5)$$

Here we introduced the initial state of the system, which transforms as irreducible representation $^1\Gamma_1^+$ of the cubic centered group O_h . The final state is the *ortho*-exciton state which transforms as $^3\Gamma_{5,xz}^+$ in Cartesian system or as $^3\Gamma_{5,1,2}^+$ in the corresponding spherical basis.

Hence, using (1, 3, 4, 5), one can deduce that the coupling of the spherical harmonic compared to the plane wave ($\hbar g_{1,2} = 124 \mu eV$) is resonantly enhanced:

$$\frac{g_{1,39}}{g_{1,2}} = -i0.06b_{1,39}(kr_0)A_{1,39}^{1,2}(r_0 + \delta r) \quad (6)$$

Here we utilized the fact that $B_{1,39}^{1,2} \ll A_{1,39}^{1,2}$. While the resonant enhancement is provided by the $b_{1,39}$ Mie coefficient here, the translational coefficient reduces the effect. That is why if one tries to couple the evanescent light to the dipole transition the effect is much weaker as $A_{1,39}^{0,1} \ll A_{1,39}^{1,2}$. The resulting exciton - evanescent light coupling is shown in Fig.1 Both dipole and quadrupole



Figure 1: The evanescent light - $1S$ quadrupole coupling ($g_{1,l}$) scaled to the bulk exciton-photon coupling ($g_{1,2}$). The size parameter kr_0 is denoted as x and the PMS is placed directly on the cuprous oxide sample ($\delta r = 0$, See also Fig.2).

coupling rate in the actual combined semiconductor-microsphere system is smaller than that in case of conventional polariton. This is attributed to the fact that the coupling occurs in a small region of the evanescent tail penetrating into cuprous oxide, although the coupling grows with mode number l , because the gradient of the evanescent field increases. Note that QE realizes *strong* coupling regime $g_{1,39} > \gamma$ while DE demonstrates *weak* regime only [7]. The property of the scalable coupling factor can be utilized in practical applications such as non-linear optics and is the subject of our future work.

3. Results and discussion

In this section let us utilize the above calculated WGM-QE interaction to obtain the evanescent polariton (EP) dispersion in the framework of the coupled oscillator model that has been

widely used for describing coupled atom-photon or exciton-photon modes in microcavity systems [13]. Near the resonance between WGM and the quadrupole exciton $\omega_{1l} \approx \omega_{1S}$ the EP branches are given by the eigenvalues of the following Hamiltonian:

$$H/\hbar = \omega_{1l} a_x^\dagger a_x + \omega_{1S} b_x^\dagger b_x + g_{1l}(x) (a_k^\dagger b_x + a_x b_x^\dagger), \quad (7)$$

here a_x, b_x are annihilation operators for light and the exciton, respectively. We also neglected kinetic energy of the QE due to smallness of the resonant wave vector and big mass of the QE. Therefore, considering that both the exciton and WGM of a single sphere are localized, the dispersion is reduced to:

$$\omega = \omega_{1S} \pm g_{1l}/\hbar \quad (8)$$

The above expression shows the formation of the doublet at resonance (both states are exactly half-QE, half-WGM). Recall that for DE-WGM weak coupling only WGM pattern shifted by the coupling has been observed [7].

The excitons are trapped in the minimum of the lower branch thus populating the strongly localized states. Physically this means that the resonant coupling with localized WGM does not let QE escape by means of its kinetic energy.

The dispersion above is similar to the quadrupole-dipole hybrid in the organic-inorganic hetero-structures [2]. In the later case, the excited organic molecules create an evanescent field penetrating into the cuprous oxide.

Now let us consider possible application of the evanescent polariton to BEC. The problem of the conventional polariton escaping from the crystal mentioned in the introduction no longer exists for the localized states of the evanescent polariton.

The linewidth of the mixed state is expected to be in the first approximation the sum of the exciton and photon linewidth weighted by the exciton and photon component of the polariton. At resonance, the linewidth of the two eigenstates is simply given by $\gamma = (\gamma_{1S} + \gamma_{m,l})/2$. Where γ_{1S} and $\gamma_{m,l}$ are the QE and WGM linewidth. Even having taken into account decrease of the Q factor due to PMS contact with the cuprous oxide sample $\gamma_{1S} \gg \gamma_{1,39}$ for the ortho-exciton. Hence, γ is defined by the QE linewidth.

For the Cu_2O para-exciton the linewidth of the polariton is given by the WGM linewidth. The para-excitons can acquire some oscillator strength provided the PMS exerts a local stress upon the cuprous oxide sample. The applied stress changes the crystal symmetry, so that usually optically inactive para-excitons may couple to the WGM. Note that due to spin-orbit interaction the para-excitons are $12 \mu eV$ below the ortho-exciton. Therefore for given radius of the PMS the resonant interaction occurs with different WGM for ortho- and para- cases.

The evanescent polariton provided by a single sphere gives the time coherence necessary for the observable BEC of the quadrupole exciton. But the spatial coherence is limited to a small region near the sphere. To improve the spatial coherence one has to sacrifice the temporal coherence slightly by delocalizing the corresponding WGM. It can be done by using an array of spheres aligned along the z direction and separated by the distance δr_0 (See Fig.2).

Recent experimental [14] and theoretical [15] studies have shown that the WGM can travel along the chain as "heavy photons". Therefore the WGM acquires the spatial dispersion, and the evanescent quadrupole polariton has the form (See Fig.3):

$$2\omega = \omega_{1l,k} + \omega_{1S} \pm \sqrt{(\omega_{1l,k} - \omega_{1S})^2 + 4|g_{1l}/\hbar|^2}$$

$$\omega_{1l,k} = \omega_{1S} + 2 \left(g_{1l}^l/\hbar \right) \cos(x - x_{1l} + \pi/2) \quad (9)$$

5



Figure 2: Schematic of formation of the evanescent polariton on linear chain of PMS. The actual dispersion is determined by the ratio of two coupling parameters such as exciton-WGM coupling and WGM-WGM coupling between the microspheres.

Here $g_{1l}^{ll} = \omega_{1S} b_{1l} A_{1l}^{ll}(\delta r_1)$ is the nearest-neighbor inter-sphere coupling parameter.



Figure 3: Dispersion of the evanescent polariton 39TE1. The dashed line (1) corresponds to the dispersion of the chain of spheres touching each other ($\delta r_0 = 0$). The thin solid line (3) stands for upper and lower branches of a single sphere dispersion ($\delta r_0 \gg \delta r = 0$). The thick solid curve (2) is the case of linear chain of the spheres in contact with the cuprous oxide ($\delta r_0 = \delta r = 0$).

When the coupling between spheres dominates ($\delta r \gg \delta r_0$) the minimum of the lower polariton branch disappears. Consequently, for possible BEC of the evanescent polariton one has to keep the desired balance between spatial and temporal coherence by adjusting experimental parameters δr and δr_0 .

Both, the energy of the $1S$ quadrupole exciton and the WGM depend on the temperature. Therefore one can use a standard temperature scan to reveal the evanescent polariton dispersion [16].

In summary, we note that there is some similarity between BEC of alkali atoms trapped by the laser field [17] and the long living QE localized by the resonant WGM.

The theory developed above is applicable also for void cavities, spherical impurities and metallic droplets in bulk cuprous oxide crystal.

4. Appendix

In the appendix we list explicit expression for the Mie scattering coefficient:

$$\begin{aligned} a_{ml} &= \frac{n^2 j_{ml}(nx) [x j_{ml}(x)]' - j_{ml}(x) [nx j_{ml}(nx)]'}{n^2 j_{ml}(nx) [x h_{ml}^{(1)}(x)]' - h_{ml}^{(1)}(x) [nx j_{ml}(nx)]'} \\ b_{ml} &= \frac{j_{ml}(nx) [x j_{ml}(x)]' - n^2 j_{ml}(x) [nx j_{ml}(nx)]'}{j_{ml}(nx) [x h_{ml}^{(1)}(x)]' - n^2 h_{ml}^{(1)}(x) [nx j_{ml}(nx)]'} \end{aligned}$$

Here $n = \epsilon^2$ is the refractive index of the spheres; $x = kr_0$ is the size parameter; j_{ml} , h_{ml} are the spherical Bessel and Hankel of the first kind functions respectively.

In the case of $l \gg 1$ the calculation of the translational coefficients can be significantly simplified with the help of the so-called maximum term approximation [9].

$$\begin{aligned} A_l' &\cong -2l(-1)^{l+1} \sqrt{\frac{l+l'}{\pi(l'+1)(l-1)}} \times \\ &\quad \frac{l'(l')'}{(l'+1)^{l'+1} (l-1)^{l-1}} h_{l+l'}^{(1)}(\eta x) \\ B_l' &\cong i \frac{x|i-j|}{ll'} A_l' \end{aligned}$$

Here η defined as $\eta = |r_0 + \delta r|/r_0 \geq 1$ is a dimensionless distance between the centers of the spheres.

References

- [1] G. Kavoulakis and G. Baym, Phys. Rev. B **53**, 7227 (1996).
- [2] O. Roslyak and J. Birman, arXiv:cond-mat/0703650, PRB to be published (2007).
- [3] D. Frohlich, G. Dasbach, G. B. Hogerthal, M. Bayer, R. Kliebera, D. Sutura, and H. Stolz, Solid State Communications **134**, 139 (2005).
- [4] C. Ell, A. L. Ivanov, and H. Haug, Phys. Rev. B **57**, 9663 (1998).
- [5] D. Snoke, Science **298**, 1368 (2002).
- [6] J. Kasprzak, M. Richard, S. Kundermann, A. Baas, P. Jeambrun, J. Keeling, F. Marchetti, M. Szymanska, R. Andre, J. Staehli, et al., Nature **443**, 409 (2006).
- [7] H. W. Xudong Fan, Scott Lacey, Optics Letters **24**, 771 (1999).
- [8] X. Fan, S. Lacey, and H. Wang, Opt. Lett **24**, 771 (1999).
- [9] H. Miyazaki and Y. Jimba, Phys. Rev. B **62**, 7976 (2000).
- [10] C. Bohren and D. Huffman, *Absorption and scattering of light by small particles* (Wiley New York, 1983).
- [11] S. Stein, Q. appl. Math **19**, 15 (1961).
- [12] K. Fuller, Appl. Opt **30**, 4716 (1991).
- [13] H. J. Carmichael, Phys. Rev. A **33**, 3262 (1986).
- [14] Y. Hara, T. Mukaiyama, K. Takeda, and M. Kuwata-Gonokami, Physical Review Letters **94**, 203905 (2005).
- [15] L. Deych and A. Roslyak, Physical Review E **73**, 36606 (2006).
- [16] E. Peter, P. Senellart, D. Martrou, A. Lemaître, J. Hours, J. Gérard, and J. Bloch, Physical Review Letters **95**, 67401 (2005).
- [17] A. J. Leggett, Rev. Mod. Phys. **73**, 307 (2001).

▶ Introduction	▶ Major Differences
▶ Installation	▶ Usage
▶ Frontmatter	▶ Floats
▶ Theorem and ...	▶ Enumerated ...
▶ Cross-references	▶ Mathematical ...
▶ Bibliography	▶ Graphical ...
▶ Final print	

1. Introduction

elsarticle.cls is a thoroughly re-written document class for formatting L^AT_EX submissions to Elsevier journals. The class uses the environments and commands defined in L^AT_EX kernel without any change in the signature so that clashes with other contributed L^AT_EX packages such as hyperref.sty, preview-latex.sty, etc., will be minimal. elsarticle.cls is primarily built upon the default article.cls. This class depends on the following packages for its proper functioning:

1. natbib.sty for citation processing;
2. geometry.sty for margin settings;
3. fleqn.clo for left aligned equations;
4. graphicx.sty for graphics inclusion;
5. txfonts.sty optional font package, if the document is to be formatted with Times and compatible math fonts;
6. hyperref.sty optional packages if hyperlinking is required in the document;
7. endfloat.sty optional packages if floats to be placed at end of the PDF.

All the above packages (except some optional packages) are part of any standard L^AT_EX installation. Therefore, the users need not be bothered about downloading any extra packages. Furthermore, users are free to make use of AMS math packages such as amsmath.sty, amsthm.sty, amssymb.sty, amsfonts.sty, etc., if they want to. All these packages work in tandem with elsarticle.cls without any problems.

2. Major Differences

Following are the major differences between elsarticle.cls and its predecessor package, elsart.cls:

- elsarticle.cls is built upon article.cls while elsart.cls is not. elsart.cls redefines many of the commands in the L^AT_EX classes/kernel, which can possibly cause surprising clashes with other contributed L^AT_EX packages;
- provides preprint document formatting by default, and optionally formats the document as per the final style of models 1+, 3+ and 5+ of Elsevier journals;

▶ Introduction	▶ Major Differences
▶ Installation	▶ Usage
▶ Frontmatter	▶ Floats
▶ Theorem and ...	▶ Enumerated ...
▶ Cross-references	▶ Mathematical ...
▶ Bibliography	▶ Graphical ...
▶ Final print	

- some easier ways for formatting `list` and `theorem` environments are provided while people can still use `amsthm.sty` package;
- `natbib.sty` is the main citation processing package which can comprehensively handle all kinds of citations and works perfectly with `hyperref.sty` in combination with `hypernat.sty`;
- long title pages are processed correctly in preprint and final formats.

3. Installation

The package is available at author resources page at Elsevier (<http://www.elsevier.com/locate/latex>). It can also be found in Comprehensive T_EX Archive Network (CTAN), <https://ctan.org/pkg/elsarticle>.

Please download the `elsarticle.dtx` which is a composite class with documentation and `elsarticle.ins` which is the L^AT_EX installer file. When we compile the `elsarticle.ins` with L^AT_EX it provides the class file, `elsarticle.cls` by stripping off all the documentation from the `*.dtx` file. The class may be moved or copied to a place, usually, `$TEXMF/tex/latex/elsevier/`, or a folder which will be read by L^AT_EX during document compilation. The T_EX file database needs updation after moving/copying class file. Usually, we use commands like `mktexlsr` or `texhash` depending upon the distribution and operating system.

4. Usage

The class should be loaded with the command:

```
\documentclass[<options>]{elsarticle}
```

where the `options` can be the following:

`preprint` default option which format the document for submission to Elsevier journals.

`review` similar to the `preprint` option, but increases the baselineskip to facilitate easier review process.

`1p` formats the article to the look and feel of the final format of model 1+ journals. This is always single column style.

`3p` formats the article to the look and feel of the final format of model 3+ journals. If the journal is a two column model, use `twocolumn` option in combination.

▶ Introduction	▶ Major Differences
▶ Installation	▶ Usage
▶ Frontmatter	▶ Floats
▶ Theorem and ...	▶ Enumerated ...
▶ Cross-references	▶ Mathematical ...
▶ Bibliography	▶ Graphical ...
▶ Final print	

`5p` formats for model 5+ journals. This is always of two column style.

`authoryear` author-year citation style of `natbib.sty`. If you want to add extra options of `natbib.sty`, you may use the options as comma delimited strings as arguments to `\biboptions` command. An example would be:

```
\biboptions{longnamesfirst,angle,semicolon}
```

`number` numbered citation style. Extra options can be loaded with `\biboptions` command.

`sort&compress` sorts and compresses the numbered citations. For example, citation [1,2,3] will become [1–3].

`longtitle` if front matter is unusually long, use this option to split the title page across pages with the correct placement of title and author footnotes in the first page.

`times` loads `txfonts.sty`, if available in the system to use Times and compatible math fonts.

`reversenotenum` Use alphabets as author–affiliation linking labels and use numbers for author footnotes. By default, numbers will be used as author–affiliation linking labels and alphabets for author footnotes.

`lefttitle` To move title and author/affiliation block to flushleft. `centertitle` is the default option which produces center alignment.

`endfloat` To place all floats at the end of the document.

`nonatbib` To unload `natbib.sty`.

`doubleblind` To hide author name, affiliation, email address etc. for double blind refereeing purpose.

All options of `article.cls` can be used with this document class.

The default options loaded are `a4paper`, `10pt`, `oneside`, `onecolumn` and `preprint`.

► Introduction	► Major Differences
► Installation	► Usage
► Frontmatter	► Floats
► Theorem and ...	► Enumerated ...
► Cross-references	► Mathematical ...
► Bibliography	► Graphical ...
► Final print	

5. Frontmatter

There are two types of frontmatter coding:

- (1) each author is connected to an affiliation with a footnote marker; hence all authors are grouped together and affiliations follow;
- (2) authors of same affiliations are grouped together and the relevant affiliation follows this group.

An example of coding the first type is provided below.

```
\title{This is a specimen title\tnoteref{t1,t2}}
\tnotetext[t1]{This document is the results of the research
  project funded by the National Science Foundation.}
\tnotetext[t2]{The second title footnote which is a longer
  text matter to fill through the whole text width and
  overflow into another line in the footnotes area of the
  first page.}
```

```
\author[1]{Jos Migchielsen\corref{cor1}%
  \fnref{fn1}}
\ead{J.Migchielsen@elsevier.com}

\author[2]{CV Radhakrishnan\fnref{fn2}}
\ead{cvr@sayahna.org}

\author[3]{CV Rajagopal\fnref{fn1,fn3}}
\ead[url]{www.stmdocs.in}

\cortext[cor1]{Corresponding author}
\fntext[fn1]{This is the first author footnote.}
\fntext[fn2]{Another author footnote, this is a very long
  footnote and it should be a really long footnote. But this
  footnote is not yet sufficiently long enough to make two
  lines of footnote text.}
\fntext[fn3]{Yet another author footnote.}
```

- | | |
|--------------------|---------------------|
| ► Introduction | ► Major Differences |
| ► Installation | ► Usage |
| ► Frontmatter | ► Floats |
| ► Theorem and ... | ► Enumerated ... |
| ► Cross-references | ► Mathematical ... |
| ► Bibliography | ► Graphical ... |
| ► Final print | |

```
\affiliation[1]{organization={Elsevier B.V.},
               addressline={Radarweg 29},
               postcode={1043 NX},
               city={Amsterdam},
               country={The Netherlands}}

\affiliation[2]{organization={Sayahna Foundation},
               addressline={JWRA 34, Jagathy},
               city={Trivandrum},
               postcode={695014},
               country={India}}

\affiliation[3]{organization={STM Document Engineering
                           Pvt Ltd.},
               addressline={Mepukada, Malayinkil},
               city={Trivandrum},
               postcode={695571},
               country={India}}
```

The output of the above T_EX source is given in Clips 1 and 2. The header portion or title area is given in Clip 1 and the footer area is given in Clip 2.

Clip 1: Header of the title page..

This is a specimen a_b title^{*,**}

Jos Migchielsen^{a,1,*}, CV Radhakrishnan^{b,2}, CV Rajagopal^{c,1,3}

^aElsevier B.V., Radarweg 29, 1043 NX Amsterdam, The Netherlands

^bSayahna Foundations, JWRA 34, Jagathy, Trivandrum 695014, India

^cSTM Document Engineering Pvt Ltd., Mepukada, Malayinkil, Trivandrum 695571, India

- | | |
|--------------------|---------------------|
| ► Introduction | ► Major Differences |
| ► Installation | ► Usage |
| ► Frontmatter | ► Floats |
| ► Theorem and ... | ► Enumerated ... |
| ► Cross-references | ► Mathematical ... |
| ► Bibliography | ► Graphical ... |
| ► Final print | |

Clip 2: Footer of the title page..

*This document is the results of the research project funded by the National Science Foundation.
**The second title footnote which is a longer text matter to fill through the whole text width and overflow into another line in the footnotes area of the first page.
*Corresponding author
Email addresses: J.Migchielsen@elsevier.com (Jos Migchielsen), cvr@sayahna.org (CV Radhakrishnan)
URL: www.stmdocs.in (CV Rajagopal)
¹This is the first author footnote.
²Another author footnote, this is a very long footnote and it should be a really long footnote. But this footnote is not yet sufficiently long enough to make two lines of footnote text.
³Yet another author footnote.

Most of the commands such as `\title`, `\author`, `\affiliation` are self explanatory. Various components are linked to each other by a label-reference mechanism; for instance, title footnote is linked to the title with a footnote mark generated by referring to the `\label` string of the `\tnotetext`. We have used similar commands such as `\tnoteref` (to link title note to title); `\corref` (to link corresponding author text to corresponding author); `\fnref` (to link footnote text to the relevant author names). $\TeX$ needs two compilations to resolve the footnote marks in the preamble part. Given below are the syntax of various note marks and note texts.

```
\tnoteref{<label(s)>}
\corref{<label(s)>}
\fnref{<label(s)>}
\tnotetext[<label>]{<title note text>}
\cortext[<label>]{<corresponding author note text>}
\fntext[<label>]{<author footnote text>}
```

where `<label(s)>` can be either one or more comma delimited label strings. The optional arguments to the `\author` command holds the ref label(s) of the address(es) to which the author is affiliated while each `\affiliation` command can have an optional argument of a label. In the same manner, `\tnotetext`, `\fntext`, `\cortext` will have optional arguments as their respective labels and note text as their mandatory argument.

The following example code provides the markup of the second type of author-affiliation.

- | | |
|--------------------|---------------------|
| ► Introduction | ► Major Differences |
| ► Installation | ► Usage |
| ► Frontmatter | ► Floats |
| ► Theorem and ... | ► Enumerated ... |
| ► Cross-references | ► Mathematical ... |
| ► Bibliography | ► Graphical ... |
| ► Final print | |

```
\author{Jos Migchielsen\corref{cor1}%
\fnref{fn1}}
\ead{J.Migchielsen@elsevier.com}
\affiliation[1]{organization={Elsevier B.V.},
addressline={Radarweg 29},
postcode={1043 NX},
city={Amsterdam},
country={The Netherlands}}

\author{CV Radhakrishnan\fnref{fn2}}
\ead{cvr@sayahna.org}
\affiliation[2]{organization={Sayahna Foundation},
addressline={JWRA 34, Jagathy},
city={Trivandrum}
postcode={695014},
country={India}}

\author{CV Rajagopal\fnref{fn1,fn3}}
\ead[url]{www.stmdocs.in}
\affiliation[3]{organization={STM Document Engineering
Pvt Ltd.},
addressline={Mepukada, Malayinkil},
city={Trivandrum}
postcode={695571},
country={India}}
```

```
\cortext[cor1]{Corresponding author}
\fnref{fn1}{This is the first author footnote.}
\fnref{fn2}{Another author footnote, this is a very long
footnote and it should be a really long footnote. But this
footnote is not yet sufficiently long enough to make two lines
of footnote text.}
```

The output of the above T_EX source is given in Clip 3.

- | | |
|--------------------|---------------------|
| ► Introduction | ► Major Differences |
| ► Installation | ► Usage |
| ► Frontmatter | ► Floats |
| ► Theorem and ... | ► Enumerated ... |
| ► Cross-references | ► Mathematical ... |
| ► Bibliography | ► Graphical ... |
| ► Final print | |

Clip 3: Header of the title page...

This is a specimen a_b title^{*,**}

Jos Migchielsen^{1,*}

Elsevier B.V., Radarweg 29, 1043 NX Amsterdam, The Netherlands

CV Radhakrishnan²

Sayahna Foundations, JWRA 34, Jagathy, Trivandrum 695014, India

CV Rajagopal^{1,3}

STM Document Engineering Pvt Ltd., Mepukada, Malayinkil, Trivandrum 695571, India

Clip 4 shows the output after giving `doubleblind` class option.

Clip 4: Double blind article.

This is a specimen a_b title

Abstract

In this work we demonstrate a_b the formation Y.1 of a new type of polariton

The frontmatter part has further environments such as abstracts and keywords. These can be marked up in the following manner:

```
\begin{abstract}
  In this work we demonstrate the formation of a new type of
  polariton on the interface between a ....
\end{abstract}
```

```
\begin{keyword}
  quadruple exciton \sep polariton \sep WGM
\end{keyword}
```

Each keyword shall be separated by a `\sep` command. msc classifications shall be provided in the keyword environment with the commands `\MSC`. `\MSC` accepts an optional argument to accommodate future revisions. eg., `\MSC[2008]`. The default is 2000.

► Introduction	► Major Differences
► Installation	► Usage
► Frontmatter	► Floats
► Theorem and ...	► Enumerated ...
► Cross-references	► Mathematical ...
► Bibliography	► Graphical ...
► Final print	

5.1. New page

Sometimes you may need to give a page-break and start a new page after title, author or abstract. Following commands can be used for this purpose.

```
\newpageafter{title}  
\newpageafter{author}  
\newpageafter{abstract}
```

`\newpageafter{title}` typeset the title alone on one page.

`\newpageafter{author}` typeset the title and author details on one page.

`\newpageafter{abstract}` typeset the title, author details and abstract & keywords one one page.

6. Floats

Figures may be included using the command, `\includegraphics` in combination with or without its several options to further control graphic. `\includegraphics` is provided by `graphic[s,x].sty` which is part of any standard L^AT_EX distribution. `graphicx.sty` is loaded by default. L^AT_EX accepts figures in the postscript format while pdfL^AT_EX accepts *.pdf, *.mps (metapost), *.jpg and *.png formats. pdfL^AT_EX does not accept graphic files in the postscript format.

The `table` environment is handy for marking up tabular material. If users want to use `multirow.sty`, `array.sty`, etc., to fine control/enhance the tables, they are welcome to load any package of their choice and `elsarticle.cls` will work in combination with all loaded packages.

7. Theorem and theorem like environments

`elsarticle.cls` provides a few shortcuts to format theorems and theorem-like environments with ease. In all commands the options that are used with the `\newtheorem` command will work exactly in the same manner. `elsarticle.cls` provides three commands to format theorem or theorem-like environments:

- | | |
|--------------------|---------------------|
| ► Introduction | ► Major Differences |
| ► Installation | ► Usage |
| ► Frontmatter | ► Floats |
| ► Theorem and ... | ► Enumerated ... |
| ► Cross-references | ► Mathematical ... |
| ► Bibliography | ► Graphical ... |
| ► Final print | |

```
\newtheorem{thm}{Theorem}
\newtheorem{lem}[thm]{Lemma}
\newdefinition{rmk}{Remark}
\newproof{pf}{Proof}
\newproof{pot}{Proof of Theorem \ref{thm2}}
```

The `\newtheorem` command formats a theorem in L^AT_EX's default style with italicized font, bold font for theorem heading and theorem number at the right hand side of the theorem heading. It also optionally accepts an argument which will be printed as an extra heading in parentheses.

```
\begin{thm}
For system (8), consensus can be achieved with
 $\|T_{\omega z}\|_{\infty} < \gamma$ 
...
\begin{eqnarray}\label{10}
....
\end{eqnarray}
\end{thm}
```

Clip 5 will show you how some text enclosed between the above code looks like:

Clip 5: `\newtheorem`.

Theorem 2. For system (8), consensus can be achieved with $\|T_{\omega z}(s)\|_{\infty} < \gamma$ if there exist a symmetric positive definite matrix $P \in \mathcal{R}^{(n-1) \times (n-1)}$ and a scalar $\mu > 0$ satisfying

$$\Gamma = \begin{bmatrix} -\bar{L}^T P - P\bar{L} + U_1^T U_1 + \mu \bar{E} & P U_1^T E_1 & P U_1^T \\ E_1^T U_1 P & -\mu I & 0 \\ U_1 P & 0 & -\gamma^2 I \end{bmatrix} < 0, \quad (10)$$

where $\bar{L} = U_1^T L U_1$ and $\bar{E} = U_1^T E_2^T E_2 U_1$.

The `\newdefinition` command is the same in all respects as its `\newtheorem` counterpart except that the font shape is roman instead of italic. Both `\newdefinition` and `\newtheorem` commands automatically define counters for the environments defined.

- | | |
|--------------------|---------------------|
| ► Introduction | ► Major Differences |
| ► Installation | ► Usage |
| ► Frontmatter | ► Floats |
| ► Theorem and ... | ► Enumerated ... |
| ► Cross-references | ► Mathematical ... |
| ► Bibliography | ► Graphical ... |
| ► Final print | |

Clip 6: `\newdefinition`.

Remark 3. We remark that; when the ratio h/λ tends to 0, the expression $\lambda L(r, s) = -(s-r)/(4\left(\frac{h}{\lambda}\right)^2 + (r-s)^2)$ tends to $1/(r-s)$ which is a singular function. This means that the expression $\lambda L(r, s)$ is not well behaved for the small values of h/λ . Consequently, for the solution to converge, the integrals of (10) and (11) must be evaluated with a large number of nodes. In our numerical applications (cf. section 5), we use 100 nodes to evaluate these integrals. With the smallest value of $h/\lambda = 0.02$, the convergence is good with $N = 20$.

The `\newproof` command defines proof environments with upright font shape. No counters are defined.

Clip 7: `\newproof`.

Proof of Theorem 2. Proof follows straightforward from Lemma 3 and Theorem 1. However, it should be emphasized that all possible $\tilde{L}_{\sigma(t)}$ should share a common Lyapunov function $V(\delta) = \delta^T(t)P\delta(t)$ (see the proof of Lemma 3 in Appendix A). $\square$

Users can also make use of `amsthm.sty` which will override all the default definitions described above.

8. Enumerated and Itemized Lists

elsarticle.cls provides an extended list processing macros which makes the usage a bit more user friendly than the default L^AT_EX list macros. With an optional argument to the `\begin{enumerate}` command, you can change the list counter type and its attributes.

```
\begin{enumerate}[1.]
\item The enumerate environment starts with an optional
      argument '1.', so that the item counter will be suffixed
      by a period.
\item You can use 'a' for alphabetical counter and '(i)' for
      roman counter.
\begin{enumerate}[a]
\item Another level of list with alphabetical counter.
\item One more item before we start another.
```

- | | |
|--------------------|---------------------|
| ► Introduction | ► Major Differences |
| ► Installation | ► Usage |
| ► Frontmatter | ► Floats |
| ► Theorem and ... | ► Enumerated ... |
| ► Cross-references | ► Mathematical ... |
| ► Bibliography | ► Graphical ... |
| ► Final print | |

Clip 8: List – Enumerate.

1. The enumerate environment starts with an optional argument '1.' so that the item counter will be suffixed by a period.
2. You can use '(a)' for alphabetical counter and '(i)' for roman counter.
 - a) Another level of list with alphabetical counter.
 - b) One more item before we start another.
 - (i) This item has roman numeral counter.
 - (ii) Another one before we close the third level.
 - c) Third item in second level.
3. All list items conclude with this step.

Further, the enhanced list environment allows one to prefix a string like 'step' to all the item numbers.

```
\begin{enumerate}[Step 1.]
\item This is the first step of the example list.
\item Obviously this is the second step.
\item The final step to wind up this example.
\end{enumerate}
```

Clip 9: List – enhanced.

- Step 1. This is the first step of the example list.
- Step 2. Obviously this is the second step.
- Step 3. The final step to wind up this example.

9. Cross-references

In electronic publications, articles may be internally hyperlinked. Hyperlinks are generated from proper cross-references in the article. For example, the words Fig. 1 will never be more than simple text, whereas the proper cross-reference `\ref{tiger}` may be turned into a hyperlink to the figure itself: [Fig. 1](#). In the same way, the words Ref. [1] will fail to turn into a hyperlink; the proper cross-reference is `\cite{Knuth96}`. Cross-referencing is possible in L^AT_EX for sections, subsections, formulae, figures, tables, and literature references.

► Introduction	► Major Differences
► Installation	► Usage
► Frontmatter	► Floats
► Theorem and ...	► Enumerated ...
► Cross-references	► Mathematical ...
► Bibliography	► Graphical ...
► Final print	

10. Mathematical symbols and formulae

Many physical/mathematical sciences authors require more mathematical symbols than the few that are provided in standard L^AT_EX. A useful package for additional symbols is the `amssymb` package, developed by the American Mathematical Society. This package includes such oft-used symbols as $\lesssim$ (`\lessssim`), $\gtrsim$ (`\gtrsim`) or $\hbar$ (`\hbar`). Note that your T_EX system should have the `msam` and `msbm` fonts installed. If you need only a few symbols, such as $\square$ (`\Box`), you might try the package `latexsym`.

Another point which would require authors' attention is the breaking up of long equations. When you use `elsarticle.cls` for formatting your submissions in the `preprint` mode, the document is formatted in single column style with a text width of 384pt or 5.3in. When this document is formatted for final print and if the journal happens to be a double column journal, the text width will be reduced to 224pt at for 3+ double column and 5+ journals respectively. All the nifty fine-tuning in equation breaking done by the author goes to waste in such cases. Therefore, authors are requested to check this problem by typesetting their submissions in final format as well just to see if their equations are broken at appropriate places, by changing appropriate options in the document class loading command, which is explained in section 4, `Usage`. This allows authors to fix any equation breaking problem before submission for publication. `elsarticle.cls` supports formatting the author submission in different types of final format. This is further discussed in section 13, `Final print`.

Displayed equations and double column journals

Many Elsevier journals print their text in two columns. Since the preprint layout uses a larger line width than such columns, the formulae are too wide for the line width in print. Here is an example of an equation (see equation 6) which is perfect in a single column preprint format:

In normal course, articles are prepared and submitted in single column format even if the final printed article will come in a double column format journal. Here the problem is that when the article is typeset by the typesetters for paginating and fit within the single column width, they have to break the lengthy equations and align them properly. Even if most of the tasks in preparing your proof is automated, the equation breaking and aligning requires manual judgement, hence this task is manual. When there comes a manual operation that area is error prone. Author needs to check that equation pretty well.

However if authors themselves break the equation to the single column width typesetters need not want to touch these area and the proof authors get will be without any errors.

- | | |
|--------------------|---------------------|
| ► Introduction | ► Major Differences |
| ► Installation | ► Usage |
| ► Frontmatter | ► Floats |
| ► Theorem and ... | ► Enumerated ... |
| ► Cross-references | ► Mathematical ... |
| ► Bibliography | ► Graphical ... |
| ► Final print | |

Clip 10: See equation (6).

Here e, m are the electron charge and mass; $\mathbf{p}$ is the electron momentum. For the quadrupole $1S$ transition in cuprous oxide the energy of interaction can be written as:

$$\sum_{i=0}^{\infty} A_n \int dx \frac{F_n(x)}{A_n + B_n} = B^n C^n \int dx \int dy \frac{G_n(x, y)}{A_n x + B_n y} + \frac{G_n(x, y)}{A_n x + B_n y} \quad (5)$$

Here we introduced the initial state of the system, which transforms as irreducible representation $^1\Gamma_1^+$ of the cubic centered group O_h . The final state is the *ortho*-exciton state which transforms as $^3\Gamma_{5,xz}^+$ in Cartesian system or as $^3\Gamma_{5,1,2}^+$ in the corresponding spherical basis.

Hence, using (1, 3, 4, 5), one can deduce that the coupling of the spherical harmonic compared to the plane wave ($\hbar g_{1,2} = 124 \mu eV$) is resonantly enhanced:

$$\frac{g_{1,39}}{g_{1,2}} = -i0.06b_{1,39}(kr_0)A_{1,39}^{1,2}(r_0 + \delta r) \quad (6)$$

Here we utilized the fact that $B_{1,39}^{1,2} \ll A_{1,39}^{1,2}$. While the resonant enhancement is provided by the $b_{1,39}$ Mie coefficient here, the translational coefficient reduces the effect. That is why if one tries to couple the evanescent light to the dipole transition the effect is much weaker as $A_{1,39}^{0,1} \ll A_{1,39}^{1,2}$.

When this document is typeset for publication in a model 3+ journal with double columns, the equation will overlap the second column text matter if the equation is not broken at the appropriate location.

Clip 11: See equation (6) overprints into second column.

Their explicit expression can be found, for instance, in [12, 9] and are explicitly listed in the Appendix.

The bulk (incident) and evanescent polaritons in cuprous oxide are formed through the quadrupole part of the light-matter interaction:

$$H_{int} = \frac{ie}{m\omega_{1S}} \mathbf{E}_{i,S} \cdot \mathbf{p}$$

Here e, m are the electron charge and mass; $\mathbf{p}$ is the electron momentum. For the quadrupole $1S$ transition in cuprous oxide the energy of interaction can be written as:

$$\sum_{i=0}^{\infty} A_n \int dx \frac{F_n(x)}{A_n + B_n} = B^n C^n \int dx \int dy \frac{G_n(x, y)}{A_n x + B_n y} + \frac{G_n(x, y)}{A_n x + B_n y} \quad (5)$$

Here we introduced the initial state of the system, which transforms as irreducible representation $^1\Gamma_1^+$ of the cubic centered group O_h . The final state is the *ortho*-exciton

Figure 1: The evanescent light - $1S$ quadrupole coupling ($g_{1,1}$) scaled to the bulk exciton-photon coupling ($g_{1,2}$). The size parameter kr_0 is denoted as x and the PMS is placed directly on the cuprous oxide sample ($\delta r = 0$, See also Fig.2).

3. Results and discussion

In this section let us utilize the above calculated WGM-QE interaction to obtain the evanescent polariton (EP) dispersion in the framework of the coupled oscillator model that has been widely used for describing coupled atom-photon or exciton-photon modes in microcavity systems [13]. Near the resonance between WGM and the quadrupole exciton $\omega_{1l} \approx \omega_{1S}$ the EP branches are given by the eigenvalues of the following Hamiltonian:

$$H/\hbar = \omega_{1l} a_x^\dagger a_x + \omega_{1S} b_x^\dagger b_x + g_{1l}(x) (a_k^\dagger b_x + a_x b_x^\dagger), \quad (7)$$

The typesetter will try to break the equation which need not necessarily be to the liking of the author or as it happens, typesetter's break point may be semantically incorrect. Therefore, authors may check their submissions for the incidence of such long equations and break the equations at the correct places so that the final typeset copy will be as they wish.

▶ Introduction	▶ Major Differences
▶ Installation	▶ Usage
▶ Frontmatter	▶ Floats
▶ Theorem and ...	▶ Enumerated ...
▶ Cross-references	▶ Mathematical ...
▶ Bibliography	▶ Graphical ...
▶ Final print	

11. Bibliography

Three bibliographic style files (*.bst) are provided — elsarticle-num.bst, elsarticle-num-names.bst and elsarticle-harv.bst — the first one can be used for the numbered scheme, second one for numbered with new options of natbib.sty. The third one is for the author year scheme.

In L^AT_EX literature, references are listed in the `\thebibliography` environment. Each reference is a `\bibitem` and each `\bibitem` is identified by a label, by which it can be cited in the text:

`\bibitem[Elson et al.(1996)]{ESG96}` is cited as `\citel{ESG96}`.

In connection with cross-referencing and possible future hyperlinking it is not a good idea to collect more than one literature item in one `\bibitem`. The so-called Harvard or author-year style of referencing is enabled by the L^AT_EX package natbib. With this package the literature can be cited as follows:

- Parenthetical: `\citep{WB96}` produces (Wettig & Brown, 1996).
- Textual: `\citel{ESG96}` produces Elson et al. (1996).
- An affix and part of a reference: `\citep[e.g.][Ch. 2]{Gea97}` produces (e.g. Governato et al., 1997, Ch. 2).

In the numbered scheme of citation, `\cite{<label>}` is used, since `\citep` or `\citel` has no relevance in the numbered scheme. natbib package is loaded by elsarticle with `numbers` as default option. You can change this to author-year or harvard scheme by adding option `authoryear` in the class loading command. If you want to use more options of the natbib package, you can do so with the `\biboptions` command, which is described in the section 4, Usage. For details of various options of the natbib package, please take a look at the natbib documentation, which is part of any standard L^AT_EX installation.

In addition to the above standard .bst files, there are 10 journal-specific .bst files also available. Instruction for using these .bst files can be found at <http://support.stmdocs.in>

12. Graphical abstract and highlights

A template for adding graphical abstract and highlights are available now. This will appear as the first two pages of the PDF before the article content begins.

- | | |
|--------------------|---------------------|
| ► Introduction | ► Major Differences |
| ► Installation | ► Usage |
| ► Frontmatter | ► Floats |
| ► Theorem and ... | ► Enumerated ... |
| ► Cross-references | ► Mathematical ... |
| ► Bibliography | ► Graphical ... |
| ► Final print | |

Please refer below to see how to code them.

```
....  
....  
  
\end{abstract}  
  
%%Graphical abstract  
\begin{graphicalabstract}  
%\includegraphics{grabs}  
\end{graphicalabstract}  
  
%%Research highlights  
\begin{highlights}  
\item Research highlight 1  
\item Research highlight 2  
\end{highlights}  
  
\begin{keyword}  
%% keywords here, in the form: keyword \sep keyword  
....  
....
```

13. Final print

The authors can format their submission to the page size and margins of their preferred journal. `elsarticle` provides four class options for the same. But it does not mean that using these options you can emulate the exact page layout of the final print copy.

1p: 1+ journals with a text area of 384pt × 562pt or 13.5cm × 19.75cm or 5.3in × 7.78in, single column style only.

3p: 3+ journals with a text area of 468pt × 622pt or 16.45cm × 21.9cm or 6.5in × 8.6in, single column style.

twocolumn: should be used along with 3p option if the journal is 3+ with the same text area as above, but double column style.

5p: 5+ with text area of 522pt × 682pt or 18.35cm × 24cm or 7.22in × 9.45in, double column style only.

Following pages have the clippings of different parts of the title page of different journal models typeset in final format.

- | | |
|--------------------|---------------------|
| ► Introduction | ► Major Differences |
| ► Installation | ► Usage |
| ► Frontmatter | ► Floats |
| ► Theorem and ... | ► Enumerated ... |
| ► Cross-references | ► Mathematical ... |
| ► Bibliography | ► Graphical ... |
| ► Final print | |

Model 1+ and 3+ will have the same look and feel in the typeset copy when presented in this document. That is also the case with the double column 3+ and 5+ journal article pages. The only difference will be wider text width of higher models. Here are the specimen single and double column journal pages.

This is a specimen a_b title^{*,**}

Jos Migchielsen^{1,*}

Elsevier B.V., Radarweg 29, 1043 NX Amsterdam, The Netherlands

CV Radhakrishnan²

Sayahna Foundations, JWRA 34, Jagathy, Trivandrum 695014, India

CV Rajagopal^{1,3}

STM Document Engineering Pvt Ltd., Mepukada, Malayinkil, Trivandrum 695571, India

Abstract

In this work we demonstrate a_b the formation Y₁ of a new type of polariton on the interface between a cuprous oxide slab and a polystyrene micro-sphere placed on the slab. The evanescent field of the resonant whispering gallery mode (WGM) of the micro sphere has a substantial gradient, and therefore effectively couples with the quadrupole 1S excitons in cuprous oxide. This evanescent polariton has a long life-time, which is determined only by its excitonic and WGM component. The polariton lower branch has a well pronounced minimum. This suggests that this excitation is localized and can be utilized for possible BEC. The spatial coherence of the polariton can be improved by assembling the micro-spheres into a linear chain.

Keywords: quadrupole exciton, polariton, WGM, BEC

JEL: 71.35.-y, 71.35.Lk, 71.36.+c

1. Introduction

Although quadrupole excitons (QE) in cuprous oxide crystals are good candidates for BEC due to their narrow line-width and long life-time there are some factors impeding BEC [1, 2]. One of these factors is that due to the small but non negligible coupling to the photon bath, one must consider BEC of the corresponding mixed light-matter states called polaritons [3]. The photon-like part of the polariton has a large group velocity and tends to escape from the crystal. Thus,

^{*}This document is the results of the research project funded by the National Science Foundation.

^{**}The second title footnote which is a longer text matter to fill through the whole text width and overflow into another line in the footnotes area of the first page.

^{*}Corresponding author

Email addresses: J.Migchielsen@elsevier.com (Jos Migchielsen), cvr@sayahna.org (CV Radhakrishnan)

URL: www.stmdocs.in (CV Rajagopal)

¹This is the first author footnote.

²Another author footnote, this is a very long footnote and it should be a really long footnote. But this footnote is not yet sufficiently long enough to make two lines of footnote text.

³Yet another author footnote.

- | | |
|--------------------|---------------------|
| ► Introduction | ► Major Differences |
| ► Installation | ► Usage |
| ► Frontmatter | ► Floats |
| ► Theorem and ... | ► Enumerated ... |
| ► Cross-references | ► Mathematical ... |
| ► Bibliography | ► Graphical ... |
| ► Final print | |

This is a specimen a_b title^{*,**}

Jos Migchielsen^{1,*}

Elsevier B.V., Radarweg 29, 1043 NX Amsterdam, The Netherlands

CV Radhakrishnan²

Sayahna Foundations, JWRA 34, Jagathy, Trivandrum 695014, India

CV Rajagopal^{1,3}

STM Document Engineering Pvt Ltd., Mepukada, Malayinkil, Trivandrum 695571, India

Abstract

In this work we demonstrate a_b the formation Y₁ of a new type of polariton on the interface between a cuprous oxide slab and a polystyrene micro-sphere placed on the slab. The evanescent field of the resonant whispering gallery mode (WGM) of the micro sphere has a substantial gradient, and therefore effectively couples with the quadrupole 1S excitons in cuprous oxide. This evanescent polariton has a long life-time, which is determined only by its excitonic and WGM component. The polariton lower branch has a well pronounced minimum. This suggests that this excitation is localized and can be utilized for possible BEC. The spatial coherence of the polariton can be improved by assembling the micro-spheres into a linear chain.

Keywords: quadrupole exciton, polariton, WGM, BEC

JEL: 71.35.-y, 71.35.Lk, 71.36.+c

1. Introduction

Although quadrupole excitons (QE) in cuprous oxide crystals are good candidates for BEC due to their narrow line-width and long life-time there are some factors impeding BEC [1, 2]. One of these factors is that due to the small but non negligible coupling to the photon bath, one must consider BEC of the corresponding mixed light-matter states called polaritons [3]. The photon-like part of the polariton has a large group velocity and tends to escape from the crystal. Thus, the temporal coherence of the condensate is effectively broken [4, 5]. One proposed solution to this issue is to place the crystal into a planar micro-cavity [6]. But even state-of-the-art planar micro-cavities can hold the light no longer than 10 μ s. Besides, formation of the polaritons in the planar cuprous oxide micro-cavity is not effective due to quadrupole origin of the excitons.

Theorem 1. *In this work we demonstrate the formation of a new type of polariton on the interface between a cuprous oxide slab and a polystyrene micro-sphere placed on the slab. The evanescent field of the resonant whispering gallery mode (WGM) of the micro sphere has a substantial gradient, and therefore effectively couples with the quadrupole 1S excitons in cuprous oxide. This evanescent polariton has a long life-time, which is determined only by its excitonic and WGM component. The polariton lower branch has a well pronounced minimum. This suggests that this excitation is localized and can be utilized for possible BEC. The spatial coherence of the polariton can be improved by assembling the micro-spheres into a linear chain.*

Therefore in this work we propose to prevent the polariton escaping by trapping it into a whispering gallery mode (WGM)⁴ of a polystyrene micro-sphere (PMS).

We develop a model which demonstrates formation of a strongly *localized* polariton-like quasi-particle. This quasi-particle is formed by the *resonant* interaction between the WGM in PMS and QE in the adjacent layer of cuprous oxide. The QE interacts with the *gradient* of the WGM evanescent field.

⁴WGM occur at particular resonant wavelengths of light for a given dielectric sphere size. At these wavelengths, the light undergoes total internal reflection at the sphere surface and becomes trapped within the particle for timescales of the order of ns .

*This document is the results of the research project funded by the National Science Foundation.

**The second title footnote which is a longer text matter to fill through the whole text width and overflow into another line in the footnotes area of the first page.

^{*}Corresponding author
Email addresses: J.Migchielsen@elsevier.com (Jos Migchielsen),
cyr@sayahna.org (CV Radhakrishnan)
URL: www.stmdocs.in (CV Rajagopal)

¹This is the first author footnote.

²Another author footnote, this is a very long footnote and it should be a really long footnote. But this footnote is not yet sufficiently long enough to make two lines of footnote text.

³Yet another author footnote.

This is a specimen a_b title^{*,**}

Jos Migchielsen^{1,*}

Elsevier B.V., Radarweg 29, 1043 NX Amsterdam, The Netherlands

CV Radhakrishnan²

Sayahna Foundations, JWRA 34, Jagathy, Trivandrum 695014, India

CV Rajagopal^{1,3}

STM Document Engineering Pvt Ltd., Mepukada, Malayinkil, Trivandrum 695571, India

Abstract

In this work we demonstrate a_b the formation Y₁ of a new type of polariton on the interface between a cuprous oxide slab and a polystyrene micro-sphere placed on the slab. The evanescent field of the resonant whispering gallery mode (WGM) of the micro sphere has a substantial gradient, and therefore effectively couples with the quadrupole 1S excitons in cuprous oxide. This evanescent polariton has a long life-time, which is determined only by its excitonic and WGM component. The polariton lower branch has a well pronounced minimum. This suggests that this excitation is localized and can be utilized for possible BEC. The spatial coherence of the polariton can be improved by assembling the micro-spheres into a linear chain.

Keywords: quadrupole exciton, polariton, WGM, BEC

JEL: 71.35.-y, 71.35.Lk, 71.36.+c

1. Introduction

Although quadrupole excitons (QE) in cuprous oxide crystals are good candidates for BEC due to their narrow line-width and long life-time there are some factors impeding BEC [1, 2]. One of these factors is that due to the small but non negligible coupling to the photon bath, one must consider BEC of the corresponding mixed light-matter states called polaritons [3]. The photon-like part of the polariton has a large group velocity and tends to escape from the crystal. Thus,

*This document is the results of the research project funded by the National Science Foundation.

**The second title footnote which is a longer text matter to fill through the whole text width and overflow into another line in the footnotes area of the first page.

*Corresponding author

Email addresses: J.Migchielsen@elsevier.com (Jos Migchielsen), cvr@sayahna.org (CV Radhakrishnan)

URL: www.stmdocs.in (CV Rajagopal)

¹This is the first author footnote.

²Another author footnote, this is a very long footnote and it should be a really long footnote. But this footnote is not yet sufficiently long enough to make two lines of footnote text.

³Yet another author footnote.

the temporal coherence of the condensate is effectively broken [4, 5]. One proposed solution to this issue is to place the crystal into a planar micro-cavity [6]. But even state-of-the-art planar micro-cavities can hold the light no longer than $10 \mu s$. Besides, formation of the polaritons in the planar cuprous oxide micro-cavity is not effective due to quadrupole origin of the excitons.

Theorem 1. *In this work we demonstrate the formation of a new type of polariton on the interface between a cuprous oxide slab and a polystyrene micro-sphere placed on the slab. The evanescent field of the resonant whispering gallery mode (WGM) of the micro sphere has a substantial gradient, and therefore effectively couples with the quadrupole $1S$ excitons in cuprous oxide. This evanescent polariton has a long life-time, which is determined only by its excitonic and WGM component. The polariton lower branch has a well pronounced minimum. This suggests that this excitation is localized and can be utilized for possible BEC. The spatial coherence of the polariton can be improved by assembling the micro-spheres into a linear chain.*

Therefore in this work we propose to prevent the polariton escaping by trapping it into a whispering gallery mode (WGM)⁴ of a polystyrene micro-sphere (PMS).

We develop a model which demonstrates formation of a strongly *localized* polariton-like quasi-particle. This quasi-particle is formed by the *resonant* interaction between the WGM in PMS and QE in the adjacent layer of cuprous oxide. The QE interacts with the *gradient* of the WGM evanescent field.

There are few experiments concerned with resonant interaction of the WGM and dipole allowed exciton (DE) [7, 8]. But the DE has some disadvantages compared to QE when it comes to interaction with the WGM. First, the evanescent light has small intensity. Therefore it is not effective for the dipole allowed coupling. But it has a large gradient, so it can effectively couple through a quadrupole part. Second, the DE has short life time compared to the QE and therefore is not suitable for BEC. Third, the kinetic energy of the DE is comparable with the interaction energy. Hence the localization is effectively impeded.

2. Evanescent vs. conventional quadrupole light-matter coupling

Assume that a *single* PMS of radius $r_0 \mu m$ is placed at a small⁵ distance $\delta r_0 \ll r_0$ from the cuprous oxide crystal ($\epsilon_{Cu_2O} = 6.5$).

There are several methods to observe WGM-QE interaction. One of them is to mount a prism (or a fiber) on the top of PMS [7]. But any surface nearby perturbs spherical symmetry and therefore reduces the Q factor and the life-time of the WGM. But the positions of the corresponding Mie resonances are not affected. Therefore in this paper we adopt a slightly different picture. Namely the scattering of the bulk polariton in cuprous oxide by the PMS. If one of the Mie resonances is in resonance with the QE one can expect formation of the new type of polariton.

Assume some density of quadrupole $1S$ excitons ([QE]), $\hbar\omega_{1S} = 2.05 eV$, $\lambda_{1S} = 2\pi/\omega_{1S} = 6096 \text{ \AA}$) has been created by an external laser pulse. The corresponding polaritons move in the crystal as the polariton and can be trapped by the PMS due to WGM-QE resonant interaction.

⁴WGM occur at particular resonant wavelengths of light for a given dielectric sphere size. At these wavelengths, the light undergoes total internal reflection at the sphere surface and becomes trapped within the particle for timescales of the order of ns .

⁵comparing to the evanescent field penetration depth

The WGM evanescent field penetration depth into the cuprous oxide adjacent crystal is much larger than the QE radius:

$$\lambda_{1S}/2\pi(\epsilon_{Cu_2O} - 1)^{1/2} = 414 \text{ \AA} \gg a_B = 4.6 \text{ \AA}$$

Hence, the light-matter interaction can be considered semi-classically. For the same reason we consider only *bulk* polaritons. However the theory may be expanded to include the surface polaritons also. In the late case the evanescent field of such a surface polariton could play an essential role in the interaction with the WGM, comparable with WGM-QE coupling.

For resonance coupling with a WGM its size parameter should be determined by the resonant wave vector in the cuprous oxide $k_0 = 2.62 \times 10^7 \text{ m}^{-1}$. For example, if one takes a polystyrene (refractive index $\epsilon^2 = 1.59$) sphere of radius $r_0 = 10.7 \text{ \mu m}$ then $k_0 r_0 = 28.78350$. This size parameter corresponds to the 39TE1 resonance [9].

The photon part of the polariton trapped inside the PMS moves as it would move in a micro-cavity of the effective modal volume $V \ll 4\pi r_0^3/3$. Consequently, it can escape through the evanescent field. This evanescent field essentially has a quantum origin and is due to tunneling through the potential caused by dielectric mismatch on the PMS surface. Therefore, we define the *evanescent* polariton (EP) as an evanescent light - QE coherent superposition.

Below we compare the evanescent quadrupole polariton and conventional bulk quadrupole polariton in cuprous oxide. For simplicity let us consider the incident polariton wave vector running along the interface (z direction). The polarization of the polariton is taken along the x direction. Therefore, in the system of coordinates centered at the sphere, the photon part of the incident polariton can be written as [10]:

$$\mathbf{E}_i = \sum_l E_0 i^l \frac{2l+1}{l(l+1)} (\mathbf{M}_{1l} - i\mathbf{N}_{1l}), \quad (1)$$

where $\mathbf{M}_{1l}$ and $\mathbf{N}_{1l}$ are vector spherical harmonics corresponding to TE- and TM- polarized modes of angular momentum l ; the z component of the angular momentum is $|m| = 1$; E_0 is the amplitude of the electric field. The scattered field is given as:

$$\mathbf{E}_s = \sum_l E_0 i^l \frac{2l+1}{l(l+1)} (ia_{1l}\mathbf{N}_{1l} - b_{1l}\mathbf{M}_{1l}), \quad (2)$$

here a_{1l} and b_{1l} are scattering Mie coefficients (See the Appendix). Taking into account that both WGM and QE have narrow line-width, and the energy separation between different WGM is much bigger than the interaction energy we adopt a single mode picture [7]. Keeping only the resonant term the last expression yields:

$$\mathbf{E}_s = -E_0 i^l 0.05 b_{1,39} \mathbf{M}_{1,39}, \quad (3)$$

To calculate the interaction of the plane wave (conventional polariton) (1) and WGM (evanescent polariton) with cuprous oxide one has to change to the cuprous oxide centered system of coordinate (See Fig.2) While in the system of the coordinate, centered at the cuprous oxide, the plane wave is still given by the expression (1), the scattered field has to be changed according to the vector spherical harmonic addition theorem [11]:

$$\mathbf{M}_{1,39} = A_{1,39}^{ml} (r_0 + \delta r) \mathbf{M}_{ml} + B_{1,39}^{ml} (r_0 + \delta r) \mathbf{N}_{ml} \quad (4)$$

Here $A_{1,39}^{ml}$ and $B_{1,39}^{ml}$ are the translational coefficients. Their explicit expression can be found, for instance, in [12, 9] and are explicitly listed in the Appendix.

The bulk (incident) and evanescent polaritons in cuprous oxide are formed through the quadrupole part of the light-matter interaction:

$$H_{int} = \frac{ie}{m\omega_{1S}} \mathbf{E}_{i,s} \cdot \mathbf{p}$$

Here e, m are the electron charge and mass; $\mathbf{p}$ is the electron momentum. For the quadrupole 1S transition in cuprous oxide the energy of interaction can be written as:

$$\hbar g = \langle {}^3\Gamma_{5,xz}^+ | H_{int} | {}^1\Gamma_1^+ \rangle = \langle {}^3\Gamma_{5,1,2}^+ | H_{int} | {}^1\Gamma_{1,0,0}^+ \rangle \quad (5)$$

Here we introduced the initial state of the system, which transforms as irreducible representation ${}^1\Gamma_1^+$ of the cubic centered group O_h . The final state is the *ortho*-exciton state which transforms as ${}^3\Gamma_{5,xz}^+$ in Cartesian system or as ${}^3\Gamma_{5,1,2}^+$ in the corresponding spherical basis.

Hence, using (1, 3, 4, 5), one can deduce that the coupling of the spherical harmonic compared to the plane wave ($\hbar g_{1,2} = 124 \mu eV$) is resonantly enhanced:

$$\frac{g_{1,39}}{g_{1,2}} = -i0.06b_{1,39}(kr_0)A_{1,39}^{1,2}(r_0 + \delta r) \quad (6)$$

Here we utilized the fact that $B_{1,39}^{1,2} \ll A_{1,39}^{1,2}$. While the resonant enhancement is provided by the $b_{1,39}$ Mie coefficient here, the translational coefficient reduces the effect. That is why if one tries to couple the evanescent light to the dipole transition the effect is much weaker as $A_{1,39}^{0,1} \ll A_{1,39}^{1,2}$. The resulting exciton - evanescent light coupling is shown in Fig.1 Both dipole and quadrupole



Figure 1: The evanescent light - 1S quadrupole coupling ($g_{1,l}$) scaled to the bulk exciton-photon coupling ($g_{1,2}$). The size parameter kr_0 is denoted as x and the PMS is placed directly on the cuprous oxide sample ($\delta r = 0$, See also Fig.2).

coupling rate in the actual combined semiconductor-microsphere system is smaller than that in case of conventional polariton. This is attributed to the fact that the coupling occurs in a small region of the evanescent tail penetrating into cuprous oxide, although the coupling grows with mode number l , because the gradient of the evanescent field increases. Note that QE realizes *strong* coupling regime $g_{1,39} > \gamma$ while DE demonstrates *weak* regime only [7]. The property of the scalable coupling factor can be utilized in practical applications such as non-linear optics and is the subject of our future work.

3. Results and discussion

In this section let us utilize the above calculated WGM-QE interaction to obtain the evanescent polariton (EP) dispersion in the framework of the coupled oscillator model that has been widely used for describing coupled atom-photon or exciton-photon modes in microcavity systems [13]. Near the resonance between WGM and the quadrupole exciton $\omega_{1l} \approx \omega_{1S}$ the EP branches are given by the eigenvalues of the following Hamiltonian:

$$H/\hbar = \omega_{1l} a_x^\dagger a_x + \omega_{1S} b_x^\dagger b_x + g_{1l}(x) (a_k^\dagger b_x + a_x b_x^\dagger), \quad (7)$$

here a_x , b_x are annihilation operators for light and the exciton, respectively. We also neglected kinetic energy of the QE due to smallness of the resonant wave vector and big mass of the QE. Therefore, considering that both the exciton and WGM of a single sphere are localized, the dispersion is reduced to:

$$\omega = \omega_{1S} \pm g_{1l}/\hbar \quad (8)$$

The above expression shows the formation of the doublet at resonance (both states are exactly half-QE, half-WGM). Recall that for DE-WGM weak coupling only WGM pattern shifted by the coupling has been observed [7].

The excitons are trapped in the minimum of the lower branch thus populating the strongly localized states. Physically this means that the resonant coupling with localized WGM does not let QE escape by means of its kinetic energy.

The dispersion above is similar to the quadrupole-dipole hybrid in the organic-inorganic hetero-structures [2]. In the later case, the excited organic molecules create an evanescent field penetrating into the cuprous oxide.

Now let us consider possible application of the evanescent polariton to BEC. The problem of the conventional polariton escaping from the crystal mentioned in the introduction no longer exists for the localized states of the evanescent polariton.

The linewidth of the mixed state is expected to be in the first approximation the sum of the exciton and photon linewidth weighted by the exciton and photon component of the polariton. At resonance, the linewidth of the two eigenstates is simply given by $\gamma = (\gamma_{1S} + \gamma_{m,l})/2$. Where γ_{1S} and $\gamma_{m,l}$ are the QE and WGM linewidth. Even having taken into account decrease of the Q factor due to PMS contact with the cuprous oxide sample $\gamma_{1S} \gg \gamma_{1,39}$ for the ortho-exciton. Hence, γ is defined by the QE linewidth.

For the Cu_2O para-exciton the linewidth of the polariton is given by the WGM linewidth. The para-excitons can acquire some oscillator strength provided the PMS exerts a local stress upon the cuprous oxide sample. The applied stress changes the crystal symmetry, so that usually optically inactive para-excitons may couple to the WGM. Note that due to spin-orbit interaction the para-excitons are $12 \mu eV$ below the ortho-exciton. Therefore for given radius of the PMS the resonant interaction occurs with different WGM for ortho- and para- cases.

The evanescent polariton provided by a single sphere gives the time coherence necessary for the observable BEC of the quadrupole exciton. But the spatial coherence is limited to a small region near the sphere. To improve the spatial coherence one has to sacrifice the temporal coherence slightly by delocalizing the corresponding WGM. It can be done by using an array of spheres aligned along the z direction and separated by the distance δr_0 (See Fig.2).

Recent experimental [14] and theoretical [15] studies have shown that the WGM can travel along the chain as "heavy photons". Therefore the WGM acquires the spatial dispersion, and the



Figure 2: Schematic of formation of the evanescent polariton on linear chain of PMS. The actual dispersion is determined by the ratio of two coupling parameters such as exciton-WGM coupling and WGM-WGM coupling between the microspheres.

evanescent quadrupole polariton has the form (See Fig.3):

$$2\omega = \omega_{1l,k} + \omega_{1S} \pm \sqrt{(\omega_{1l,k} - \omega_{1S})^2 + 4|g_{1l}/\hbar|^2}$$

$$\omega_{1l,k} = \omega_{1S} + 2\left(g_{1l}^I/\hbar\right)\cos(x - x_{1l} + \pi/2) \quad (9)$$

Here $g_{1l}^I = \omega_{1S} b_{1l} A_{1l}^I(\delta r_1)$ is the nearest-neighbor inter-sphere coupling parameter.



Figure 3: Dispersion of the evanescent polariton 39TE1. The dashed line (1) corresponds to the dispersion of the chain of spheres touching each other ($\delta r_0 = 0$). The thin solid line (3) stands for upper and lower branches of a single sphere dispersion ($\delta r_0 \gg \delta r = 0$). The thick solid curve (2) is the case of linear chain of the spheres in contact with the cuprous oxide ($\delta r_0 = \delta r = 0$).

When the coupling between spheres dominates ($\delta r \gg \delta r_0$) the minimum of the lower polariton branch disappears. Consequently, for possible BEC of the evanescent polariton one has to keep the desired balance between spatial and temporal coherence by adjusting experimental parameters δr and δr_0 .

Both, the energy of the 1S quadrupole exciton and the WGM depend on the temperature.

Therefore one can use a standard temperature scan to reveal the evanescent polariton dispersion [16].

In summary, we note that there is some similarity between BEC of alkali atoms trapped by the laser field [17] and the long living QE localized by the resonant WGM.

The theory developed above is applicable also for void cavities, spherical impurities and metallic droplets in bulk cuprous oxide crystal.

4. Appendix

In the appendix we list explicit expression for the Mie scattering coefficient:

$$\begin{aligned} a_{ml} &= \frac{n^2 j_{ml}(nx) [x j_{ml}(x)]' - j_{ml}(x) [nx j_{ml}(nx)]'}{n^2 j_{ml}(nx) [x h_{ml}^{(1)}(x)]' - h_{ml}^{(1)}(x) [nx j_{ml}(nx)]'} \\ b_{ml} &= \frac{j_{ml}(nx) [x j_{ml}(x)]' - n^2 j_{ml}(x) [nx j_{ml}(nx)]'}{j_{ml}(nx) [x h_{ml}^{(1)}(x)]' - n^2 h_{ml}^{(1)}(x) [nx j_{ml}(nx)]'} \end{aligned}$$

Here $n = \epsilon^2$ is the refractive index of the spheres; $x = kr_0$ is the size parameter; j_{ml} , h_{ml} are the spherical Bessel and Hankel of the first kind functions respectively.

In the case of $l \gg 1$ the calculation of the translational coefficients can be significantly simplified with the help of the so-called maximum term approximation [9].

$$\begin{aligned} A_l' &\cong -2l(-1)^{l+1} \sqrt{\frac{l+l'}{\pi(l'+1)(l-1)}} \times \\ &\quad \frac{l'(l')'}{(l'+1)^{l'+1} (l-1)^{l-1}} h_{l+l'}^{(1)}(\eta x) \\ B_l'' &\cong i \frac{x|i-j|}{ll'} A_l'' \end{aligned}$$

Here η defined as $\eta = |r_0 + \delta r|/r_0 \geq 1$ is a dimensionless distance between the centers of the spheres.

References

- [1] G. Kavoulakis and G. Baym, Phys. Rev. B **53**, 7227 (1996).
- [2] O. Roslyak and J. Birman, arXiv:cond-mat/0703650, PRB to be published (2007).
- [3] D. Frohlich, G. Dasbach, G. B. Hogerthal, M. Bayer, R. Kliebera, D. Suter, and H. Stolz, Solid State Communications **134**, 139 (2005).
- [4] C. Ell, A. L. Ivanov, and H. Haug, Phys. Rev. B **57**, 9663 (1998).
- [5] D. Snoke, Science **298**, 1368 (2002).
- [6] J. Kasprzak, M. Richard, S. Kundermann, A. Baas, P. Jeambrun, J. Keeling, F. Marchetti, M. Szymanska, R. Andre, J. Staehli, et al., Nature **443**, 409 (2006).
- [7] H. W. Xudong Fan, Scott Lacey, Optics Letters **24**, 771 (1999).
- [8] X. Fan, S. Lacey, and H. Wang, Opt. Lett **24**, 771 (1999).
- [9] H. Miyazaki and Y. Jimba, Phys. Rev. B **62**, 7976 (2000).
- [10] C. Bohren and D. Huffman, *Absorption and scattering of light by small particles* (Wiley New York, 1983).
- [11] S. Stein, Q. appl. Math **19**, 15 (1961).
- [12] K. Fuller, Appl. Opt **30**, 4716 (1991).
- [13] H. J. Carmichael, Phys. Rev. A **33**, 3262 (1986).

- [14] Y. Hara, T. Mukaiyama, K. Takeda, and M. Kuwata-Gonokami, Physical Review Letters **94**, 203905 (2005).
- [15] L. Deych and A. Roslyak, Physical Review E **73**, 36606 (2006).
- [16] E. Peter, P. Senellart, D. Martrou, A. Lemaître, J. Hours, J. Gérard, and J. Bloch, Physical Review Letters **95**, 67401 (2005).
- [17] A. J. Leggett, Rev. Mod. Phys. **73**, 307 (2001).

This is a specimen a_b title

Abstract

In this work we demonstrate a_b the formation Y_1 of a new type of polariton on the interface between a cuprous oxide slab and a polystyrene micro-sphere placed on the slab. The evanescent field of the resonant whispering gallery mode (WGM) of the micro sphere has a substantial gradient, and therefore effectively couples with the quadrupole $1S$ excitons in cuprous oxide. This evanescent polariton has a long life-time, which is determined only by its excitonic and WGM component. The polariton lower branch has a well pronounced minimum. This suggests that this excitation is localized and can be utilized for possible BEC. The spatial coherence of the polariton can be improved by assembling the micro-spheres into a linear chain.

Keywords: quadrupole exciton, polariton, WGM, BEC

JEL: 71.35.-y, 71.35.Lk, 71.36.+c

1. Introduction

Although quadrupole excitons (QE) in cuprous oxide crystals are good candidates for BEC due to their narrow line-width and long life-time there are some factors impeding BEC [Kavoulakis and Baym \(1996\)](#); [Roslyak and Birman \(2007\)](#). One of these factors is that due to the small but non negligible coupling to the photon bath, one must consider BEC of the corresponding mixed light-matter states called polaritons [Frohlich et al. \(2005\)](#). The photon-like part of the polariton has a large group velocity and tends to escape from the crystal.

¹This is the first author footnote.

²Another author footnote, this is a very long footnote and it should be a really long footnote. But this footnote is not yet sufficiently long enough to make two lines of footnote text.

³Yet another author footnote.

Thus, the temporal coherence of the condensate is effectively broken [Ell et al. \(1998\)](#); [Snoke \(2002\)](#). One proposed solution to this issue is to place the crystal into a planar micro-cavity [Kasprzak et al. \(2006\)](#). But even state-of-the-art planar micro-cavities can hold the light no longer than $10 \mu s$. Besides, formation of the polaritons in the planar cuprous oxide micro-cavity is not effective due to quadrupole origin of the excitons.

Theorem 1. *In this work we demonstrate the formation of a new type of polariton on the interface between a cuprous oxide slab and a polystyrene micro-sphere placed on the slab. The evanescent field of the resonant whispering gallery mode (WGM) of the micro sphere has a substantial gradient, and therefore effectively couples with the quadrupole $1S$ excitons in cuprous oxide. This evanescent polariton has a long life-time, which is determined only by its excitonic and WGM component. The polariton lower branch has a well pronounced minimum. This suggests that this excitation is localized and can be utilized for possible BEC. The spatial coherence of the polariton can be improved by assembling the micro-spheres into a linear chain.*

Therefore in this work we propose to prevent the polariton escaping by trapping it into a whispering gallery mode (WGM)⁴ of a polystyrene micro-sphere (PMS).

We develop a model which demonstrates formation of a strongly *localized* polariton-like quasi-particle. This quasi-particle is formed by the *resonant* interaction between the WGM in PMS and QE in the adjacent layer of cuprous oxide. The QE interacts with the *gradient* of the WGM evanescent field.

There are few experiments concerned with resonant interaction of the WGM and dipole allowed exciton (DE) [Xudong Fan \(1999\)](#); [Fan et al. \(1999\)](#). But the DE has some disadvantages compared to QE when it comes to interaction

⁴WGM occur at particular resonant wavelengths of light for a given dielectric sphere size. At these wavelengths, the light undergoes total internal reflection at the sphere surface and becomes trapped within the particle for timescales of the order of ns .

with the WGM. First, the evanescent light has small intensity. Therefore it is not effective for the dipole allowed coupling. But it has a large gradient, so it can effectively couple through a quadrupole part. Second, the DE has short life time compared to the QE and therefore is not suitable for BEC. Third, the kinetic energy of the DE is comparable with the interaction energy. Hence the localization is effectively impeded.

2. Evanescent vs. conventional quadrupole light-matter coupling

Assume that a *single* PMS of radius r_0 μm is placed at a small ⁵ distance $\delta r_0 \ll r_0$ from the cuprous oxide crystal ($\epsilon_{Cu_2O} = 6.5$).

There are several methods to observe WGM-QE interaction. One of them is to mount a prism (or a fiber) on the top of PMS [Xudong Fan \(1999\)](#). But any surface nearby perturbs spherical symmetry and therefore reduces the Q factor and the life-time of the WGM. But the positions of the corresponding Mie resonances are not affected. Therefore in this paper we adopt a slightly different picture. Namely the scattering of the bulk polariton in cuprous oxide by the PMS. If one of the Mie resonances is in resonance with the QE one can expect formation of the new type of polariton.

Assume some density of quadrupole $1S$ excitons ($[QE]$), $\hbar\omega_{1S} = 2.05$ eV, $\lambda_{1S} = 2\pi/\omega_{1S} = 6096$ Å) has been created by an external laser pulse. The corresponding polaritons move in the crystal as the polariton and can be trapped by the PMS due to WGM-QE resonant interaction.

The WGM evanescent field penetration depth into the cuprous oxide adjacent crystal is much larger than the QE radius:

$$\lambda_{1S}/2\pi (\epsilon_{Cu_2O} - 1)^{1/2} = 414 \text{ Å} \gg a_B = 4.6 \text{ Å}$$

Hence, the light-matter interaction can be considered semi-classically. For the same reason we consider only *bulk* polaritons. However the theory may be

⁵comparing to the evanescent field penetration depth

expanded to include the surface polaritons also. In the late case the evanescent field of such a surface polariton could play an essential role in the interaction with the WGM, comparable with WGM-QE coupling.

For resonance coupling with a WGM its size parameter should be determined by the resonant wave vector in the cuprous oxide $k_0 = 2.62 \times 10^7 \text{ m}^{-1}$. For example, if one takes a polystyrene (refractive index $\epsilon^2 = 1.59$) sphere of radius $r_0 = 10.7 \text{ }\mu\text{m}$ then $k_0 r_0 = 28.78350$. This size parameter corresponds to the 39TE1 resonance [Miyazaki and Jimba \(2000\)](#).

The photon part of the polariton trapped inside the PMS moves as it would move in a micro-cavity of the effective modal volume $V \ll 4\pi r_0^3/3$. Consequently, it can escape through the evanescent field. This evanescent field essentially has a quantum origin and is due to tunneling through the potential caused by dielectric mismatch on the PMS surface. Therefore, we define the *evanescent* polariton (EP) as an evanescent light - QE coherent superposition.

Below we compare the evanescent quadrupole polariton and conventional bulk quadrupole polariton in cuprous oxide. For simplicity let us consider the incident polariton wave vector running along the interface (z direction). The polarization of the polariton is taken along the x direction. Therefore, in the system of coordinates centered at the sphere, the photon part of the incident polariton can be written as [Bohren and Huffman \(1983\)](#):

$$\mathbf{E}_i = \sum_l E_0 i^l \frac{2l+1}{l(l+1)} (\mathbf{M}_{1l} - i\mathbf{N}_{1l}), \quad (1)$$

where $\mathbf{M}_{1l}$ and $\mathbf{N}_{1l}$ are vector spherical harmonics corresponding to TE- and TM- polarized modes of angular momentum l ; the z component of the angular momentum is $|m| = 1$; E_0 is the amplitude of the electric field. The scattered field is given as:

$$\mathbf{E}_s = \sum_l E_0 i^l \frac{2l+1}{l(l+1)} (ia_{1l}\mathbf{N}_{1l} - b_{1l}\mathbf{M}_{1l}), \quad (2)$$

here a_{1l} and b_{1l} are scattering Mie coefficients (See the Appendix). Taking into account that both WGM and QE have narrow line-width, and the energy separation between different WGM is much bigger then the interaction energy

we adopt a single mode picture [Xudong Fan \(1999\)](#). Keeping only the resonant term the last expression yields:

$$\mathbf{E}_s = -E_0 i^l 0.05 b_{1,39} \mathbf{M}_{1,39}, \quad (3)$$

To calculate the interaction of the plane wave (conventional polariton) (1) and WGM (evanescent polariton) with cuprous oxide one has to change to the cuprous oxide centered system of coordinate (See Fig.2) While in the system of the coordinate, centered at the cuprous oxide, the plane wave is still given by the expression (1), the scattered field has to be changed according to the vector spherical harmonic addition theorem [Stein \(1961\)](#):

$$\mathbf{M}_{1,39} = A_{1,39}^{ml} (r_0 + \delta r) \mathbf{M}_{ml} + B_{1,39}^{ml} (r_0 + \delta r) \mathbf{N}_{ml} \quad (4)$$

Here $A_{1,39}^{ml}$ and $B_{1,39}^{ml}$ are the translational coefficients. Their explicit expression can be found, for instance, in [Fuller \(1991\)](#); [Miyazaki and Jimba \(2000\)](#) and are explicitly listed in the Appendix.

The bulk (incident) and evanescent polaritons in cuprous oxide are formed through the quadrupole part of the light-matter interaction:

$$H_{int} = \frac{ie}{m\omega_{1S}} \mathbf{E}_{i,s} \cdot \mathbf{p}$$

Here e, m are the electron charge and mass; $\mathbf{p}$ is the electron momentum. For the quadrupole $1S$ transition in cuprous oxide the energy of interaction can be written as:

$$\hbar g = \langle {}^3\Gamma_{5,xz}^+ | H_{int} | {}^1\Gamma_1^+ \rangle = \langle {}^3\Gamma_{5,1,2}^+ | H_{int} | {}^1\Gamma_{1,0,0}^+ \rangle \quad (5)$$

Here we introduced the initial state of the system, which transforms as irreducible representation ${}^1\Gamma_1^+$ of the cubic centered group O_h . The final state is the *ortho*-exciton state which transforms as ${}^3\Gamma_{5,xz}^+$ in Cartesian system or as ${}^3\Gamma_{5,1,2}^+$ in the corresponding spherical basis.

Hence, using (1, 3, 4, 5), one can deduce that the coupling of the spherical harmonic compared to the plane wave ($\hbar g_{1,2} = 124 \mu eV$) is resonantly enhanced:

$$\frac{g_{1,39}}{g_{1,2}} = -i 0.06 b_{1,39} (kr_0) A_{1,39}^{1,2} (r_0 + \delta r) \quad (6)$$

Here we utilized the fact that $B_{1,39}^{1,2} \ll A_{1,39}^{1,2}$. While the resonant enhancement is provided by the $b_{1,39}$ Mie coefficient here, the translational coefficient reduces the effect. That is why if one tries to couple the evanescent light to the dipole transition the effect is much weaker as $A_{1,39}^{0,1} \ll A_{1,39}^{1,2}$. The resulting exciton - evanescent light coupling is shown in Fig.1 Both dipole and quadrupole coupling

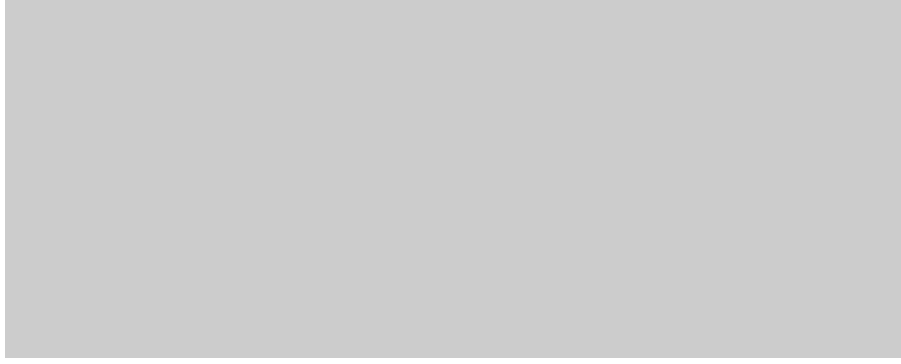


Figure 1: The evanescent light - 1S quadrupole coupling ($g_{1,l}$) scaled to the bulk exciton-photon coupling ($g_{1,2}$). The size parameter kr_0 is denoted as x and the PMS is placed directly on the cuprous oxide sample ($\delta r = 0$, See also Fig.2).

rate in the actual combined semiconductor-microsphere system is smaller than that in case of conventional polariton. This is attributed to the fact that the coupling occurs in a small region of the evanescent tail penetrating into cuprous oxide, although the coupling grows with mode number l , because the gradient of the evanescent field increases. Note that QE realizes *strong* coupling regime $g_{1,39} > \gamma$ while DE demonstrates *weak* regime only [Xudong Fan \(1999\)](#). The property of the scalable coupling factor can be utilized in practical applications such as non-linear optics and is the subject of our future work.

3. Results and discussion

In this section let us utilize the above calculated WGM-QE interaction to obtain the evanescent polariton (EP) dispersion in the framework of the coupled oscillator model that has been widely used for describing coupled atom-

photon or exciton-photon modes in microcavity systems [Carmichael \(1986\)](#). Near the resonance between WGM and the quadrupole exciton $\omega_{1l} \approx \omega_{1S}$ the EP branches are given by the eigenvalues of the following Hamiltonian:

$$H/\hbar = \omega_{1l} a_x^\dagger a_x + \omega_{1S} b_x^\dagger b_x + g_{1l}(x) \left(a_k^\dagger b_x + a_x b_x^\dagger \right), \quad (7)$$

here a_x , b_x are annihilation operators for light and the exciton, respectively. We also neglected kinetic energy of the QE due to smallness of the resonant wave vector and big mass of the QE. Therefore, considering that both the exciton and WGM of a single sphere are localized, the dispersion is reduced to:

$$\omega = \omega_{1S} \pm g_{1l}/\hbar \quad (8)$$

The above expression shows the formation of the doublet at resonance (both states are exactly half-QE, half-WGM). Recall that for DE-WGM weak coupling only WGM pattern shifted by the coupling has been observed [Xudong Fan \(1999\)](#).

The excitons are trapped in the minimum of the lower branch thus populating the strongly localized states. Physically this means that the resonant coupling with localized WGM does not let QE escape by means of its kinetic energy.

The dispersion above is similar to the quadrupole-dipole hybrid in the organic-inorganic hetero-structures [Roslyak and Birman \(2007\)](#). In the later case, the excited organic molecules create an evanescent field penetrating into the cuprous oxide.

Now let us consider possible application of the evanescent polariton to BEC. The problem of the conventional polariton escaping from the crystal mentioned in the introduction no longer exists for the localized states of the evanescent polariton.

The linewidth of the mixed state is expected to be in the first approximation the sum of the exciton and photon linewidth weighted by the exciton and photon component of the polariton. At resonance, the linewidth of the two eigenstates is simply given by $\gamma = (\gamma_{1S} + \gamma_{m,l})/2$. Where γ_{1S} and $\gamma_{m,l}$ are the QE and

WGM linewidth. Even having taken into account decrease of the Q factor due to PMS contact with the cuprous oxide sample $\gamma_{1S} \gg \gamma_{1,39}$ for the ortho-exciton. Hence, γ is defined by the QE linewidth.

For the Cu_2O para-exciton the linewidth of the polariton is given by the WGM linewidth. The para-excitons can acquire some oscillator strength provided the PMS exerts a local stress upon the cuprous oxide sample. The applied stress changes the crystal symmetry, so that usually optically inactive para-excitons may couple to the WGM. Note that due to spin-orbit interaction the para-excitons are $12 \mu eV$ below the ortho-exciton. Therefore for given radius of the PMS the resonant interaction occurs with different WGM for ortho- and para- cases.

The evanescent polariton provided by a single sphere gives the time coherence necessary for the observable BEC of the quadrupole exciton. But the spatial coherence is limited to a small region near the sphere. To improve the spatial coherence one has to sacrifice the temporal coherence slightly by delocalizing the corresponding WGM. It can be done by using an array of spheres aligned along the z direction and separated by the distance δr_0 (See Fig.2).



Figure 2: Schematic of formation of the evanescent polariton on linear chain of PMS. The actual dispersion is determined by the ratio of two coupling parameters such as exciton-WGM coupling and WGM-WGM coupling between the microspheres.

Recent experimental [Hara et al. \(2005\)](#) and theoretical [Deych and Roslyak \(2006\)](#) studies have shown that the WGM can travel along the chain as "heavy

photons". Therefore the WGM acquires the spatial dispersion, and the evanescent quadrupole polariton has the form (See Fig.3):

$$\begin{aligned} 2\omega &= \omega_{1l,k} + \omega_{1S} \pm \sqrt{(\omega_{1l,k} - \omega_{1S})^2 + 4|g_{1l}/\hbar|^2} \\ \omega_{1l,k} &= \omega_{1S} + 2 \left(g_{1l}^{\text{II}}/\hbar \right) \cos(x - x_{1l} + \pi/2) \end{aligned} \quad (9)$$

Here $g_{1l}^{\text{II}} = \omega_{1S} b_{1l} A_{1l}^{\text{II}}(\delta r_1)$ is the nearest-neighbor inter-sphere coupling parameter.



Figure 3: Dispersion of the evanescent polariton 39TE1. The dashed line (1) corresponds to the dispersion of the chain of spheres touching each other ($\delta r_0 = 0$). The thin solid line (3) stands for upper and lower branches of a single sphere dispersion ($\delta r_0 \gg \delta r = 0$). The thick solid curve (2) is the case of linear chain of the spheres in contact with the cuprous oxide ($\delta r_0 = \delta r = 0$).

When the coupling between spheres dominates ($\delta r \gg \delta r_0$) the minimum of the lower polariton branch disappears. Consequently, for possible BEC of the evanescent polariton one has to keep the desired balance between spatial and temporal coherence by adjusting experimental parameters δr and δr_0 .

Both, the energy of the 1S quadrupole exciton and the WGM depend on the temperature. Therefore one can use a standard temperature scan to reveal the evanescent polariton dispersion [Peter et al. \(2005\)](#).

In summary, we note that there is some similarity between BEC of alkali atoms trapped by the laser field [Leggett \(2008\)](#) and the long living QE localized by the resonant WGM.

The theory developed above is applicable also for void cavities, spherical

impurities and metallic droplets in bulk cuprous oxide crystal.

4. Appendix

In the appendix we list explicit expression for the Mie scattering coefficient:

$$\begin{aligned} a_{ml} &= \frac{n^2 j_{ml}(nx) [x j_{ml}(x)]' - j_{ml}(x) [nx j_{ml}(nx)]'}{n^2 j_{ml}(nx) [x h_{ml}^{(1)}(x)]' - h_{ml}^{(1)}(x) [nx j_{ml}(nx)]'} \\ b_{ml} &= \frac{j_{ml}(nx) [x j_{ml}(x)]' - n^2 j_{ml}(x) [nx j_{ml}(nx)]'}{j_{ml}(nx) [x h_{ml}^{(1)}(x)]' - n^2 h_{ml}^{(1)}(x) [nx j_{ml}(nx)]'} \end{aligned}$$

Here $n = \epsilon^2$ is the refractive index of the spheres; $x = kr_0$ is the size parameter; j_{ml} , h_{ml} are the spherical Bessel and Hankel of the first kind functions respectively.

In the case of $l \gg 1$ the calculation of the translational coefficients can be significantly simplified with the help of the so-called maximum term approximation [Miyazaki and Jimba \(2000\)](#).

$$\begin{aligned} A_l^{l'} &\cong -2l(-1)^{l+1} \sqrt{\frac{l+l'}{\pi(l'+1)(l-1)}} \times \\ &\quad \frac{l^l (l')^{l'}}{(l'+1)^{l'+1} (l-1)^{l-1}} h_{l+l'}^{(1)}(\eta x) \\ B_l^{l'} &\cong i \frac{x|i-j|}{ll'} A_l^{l'} \end{aligned}$$

Here η defined as $\eta = |r_0 + \delta r|/r_0 \geq 1$ is a dimensionless distance between the centers of the spheres.

References

- G. Kavoulakis and G. Baym, Phys. Rev. B **53**, 7227 (1996).
- O. Roslyak and J. Birman, arXiv:cond-mat/0703650, PRB to be published (2007).
- D. Frohlich, G. Dasbach, G. B. Hogerthal, M. Bayer, R. Kliebera, D. Sutura, and H. Stolz, Solid State Communications **134**, 139 (2005).
- C. Ell, A. L. Ivanov, and H. Haug, Phys. Rev. B **57**, 9663 (1998).
- D. Snoke, Science **298**, 1368 (2002).

- J. Kasprzak, M. Richard, S. Kundermann, A. Baas, P. Jeambrun, J. Keeling, F. Marchetti, M. Szymanska, R. Andre, J. Staehli, et al., *Nature* **443**, 409 (2006).
- H. W. Xudong Fan, Scott Lacey, *Optics Letters* **24**, 771 (1999).
- X. Fan, S. Lacey, and H. Wang, *Opt. Lett* **24**, 771 (1999).
- H. Miyazaki and Y. Jimba, *Phys. Rev. B* **62**, 7976 (2000).
- C. Bohren and D. Huffman, *Absorption and scattering of light by small particles* (Wiley New York, 1983).
- S. Stein, *Q. appl. Math* **19**, 15 (1961).
- K. Fuller, *Appl. Opt* **30**, 4716 (1991).
- H. J. Carmichael, *Phys. Rev. A* **33**, 3262 (1986).
- Y. Hara, T. Mukaiyama, K. Takeda, and M. Kuwata-Gonokami, *Physical Review Letters* **94**, 203905 (2005).
- L. Deych and A. Roslyak, *Physical Review E* **73**, 36606 (2006).
- E. Peter, P. Senellart, D. Martrou, A. Lemaître, J. Hours, J. Gérard, and J. Bloch, *Physical Review Letters* **95**, 67401 (2005).
- A. J. Leggett, *Rev. Mod. Phys.* **73**, 307 (2001).

This is a specimen a_b title^{*,**}

Jos Migchielsen^{a,1,*}, CV Radhakrishnan^{b,2}, CV Rajagopal^{c,1,3}

^aElsevier B.V., Radarweg 29, 1043 NX Amsterdam, The Netherlands

^bSayahna Foundations, JWRA 34, Jagathy, Trivandrum 695014, India

^cSTM Document Engineering Pvt Ltd., Mepukada, Malayinkil, Trivandrum 695571, India

Abstract

In this work we demonstrate a_b the formation Y_1 of a new type of polariton on the interface between a cuprous oxide slab and a polystyrene micro-sphere placed on the slab. The evanescent field of the resonant whispering gallery mode (WGM) of the micro sphere has a substantial gradient, and therefore effectively couples with the quadrupole 1S excitons in cuprous oxide. This evanescent polariton has a long life-time, which is determined only by its excitonic and WGM component. The polariton lower branch has a well pronounced minimum. This suggests that this excitation is localized and can be utilized for possible BEC. The spatial coherence of the polariton can be improved by assembling the micro-spheres into a linear chain.

Keywords: quadrupole exciton, polariton, WGM, BEC

JEL: 71.35.-y, 71.35.Lk, 71.36.+c

1. Introduction

Although quadrupole excitons (QE) in cuprous oxide crystals are good candidates for BEC due to their narrow line-width and long life-time there are some factors impeding BEC [1, 2]. One of these factors is that due to the small but non negligible coupling to the photon bath, one must consider BEC of the corresponding mixed light-matter states called polaritons [3]. The photon-like part of the polariton has a large group velocity and tends to escape from the crystal. Thus, the temporal coherence of the condensate is effectively broken [4, 5]. One proposed solution to this issue is to place the crystal into a planar micro-cavity [6]. But even state-of-the-art planar micro-cavities can hold the light no longer than 10 μ s. Besides, formation of the polaritons in the planar cuprous oxide micro-cavity is not effective due to quadrupole origin of the excitons.

Theorem 1. *In this work we demonstrate the formation of a new type of polariton on the interface between a cuprous oxide slab and a polystyrene micro-sphere placed on the slab. The evanescent field of the resonant whispering gallery mode (WGM) of the micro sphere has a substantial gradient, and therefore effectively couples with the quadrupole 1S excitons in cuprous oxide. This evanescent polariton has a long life-time, which is determined only by its excitonic and WGM component. The polariton lower branch has a well pronounced minimum. This suggests that this excitation is localized and can be utilized for possible BEC. The spatial coherence of the polariton can be improved by assembling the micro-spheres into a linear chain.*

*This document is the results of the research project funded by the National Science Foundation.

**The second title footnote which is a longer text matter to fill through the whole text width and overflow into another line in the footnotes area of the first page.

*Corresponding author

Email addresses: J.Migchielsen@elsevier.com (Jos Migchielsen), cvr@sayahna.org (CV Radhakrishnan)

URL: www.stmdocs.in (CV Rajagopal)

¹This is the first author footnote.

²Another author footnote, this is a very long footnote and it should be a really long footnote. But this footnote is not yet sufficiently long enough to make two lines of footnote text.

³Yet another author footnote.

Therefore in this work we propose to prevent the polariton escaping by trapping it into a whispering gallery mode (WGM)⁴ of a polystyrene micro-sphere (PMS).

We develop a model which demonstrates formation of a strongly *localized* polariton-like quasi-particle. This quasi-particle is formed by the *resonant* interaction between the WGM in PMS and QE in the adjacent layer of cuprous oxide. The QE interacts with the *gradient* of the WGM evanescent field.

There are few experiments concerned with resonant interaction of the WGM and dipole allowed exciton (DE) [7, 8]. But the DE has some disadvantages compared to QE when it comes to interaction with the WGM. First, the evanescent light has small intensity. Therefore it is not effective for the dipole allowed coupling. But it has a large gradient, so it can effectively couple through a quadrupole part. Second, the DE has short life time compared to the QE and therefore is not suitable for BEC. Third, the kinetic energy of the DE is comparable with the interaction energy. Hence the localization is effectively impeded.

2. Evanescent vs. conventional quadrupole light-matter coupling

Assume that a *single* PMS of radius r_0 μm is placed at a small⁵ distance $\delta r_0 \ll r_0$ from the cuprous oxide crystal ($\epsilon_{\text{Cu}_2\text{O}} = 6.5$).

There are several methods to observe WGM-QE interaction. One of them is to mount a prism (or a fiber) on the top of PMS [7]. But any surface nearby perturbs spherical symmetry and therefore reduces the Q factor and the life-time of the WGM. But the positions of the corresponding Mie resonances are not affected. Therefore in this paper we adopt a slightly different picture. Namely the scattering of the bulk polariton in cuprous oxide by the PMS. If one of the Mie resonances is in resonance with the QE one can expect formation of the new type of polariton.

Assume some density of quadrupole 1S excitons ([QE]), $\hbar\omega_{1S} = 2.05$ eV, $\lambda_{1S} = 2\pi/\omega_{1S} = 6096$ Å) has been created by an external laser pulse. The corresponding polaritons move in the crystal as the polariton and can be trapped by the PMS due to WGM-QE resonant interaction.

The WGM evanescent field penetration depth into the cuprous oxide adjacent crystal is much larger than the QE radius:

$$\lambda_{1S}/2\pi(\epsilon_{\text{Cu}_2\text{O}} - 1)^{1/2} = 414 \text{ Å} \gg a_B = 4.6 \text{ Å}$$

Hence, the light-matter interaction can be considered semi-classically. For the same reason we consider only *bulk* polaritons. However the theory may be expanded to include the surface polaritons also. In the late case the evanescent field of such a surface polariton could play an essential role in the interaction with the WGM, comparable with WGM-QE coupling.

For resonance coupling with a WGM its size parameter should be determined by the resonant wave vector in the cuprous oxide $k_0 = 2.62 \times 10^7 \text{ m}^{-1}$. For example, if one takes a polystyrene (refractive index $\epsilon^2 = 1.59$) sphere of radius $r_0 = 10.7 \mu\text{m}$ then $k_0 r_0 = 28.78350$. This size parameter corresponds to the 39TE1 resonance [9].

The photon part of the polariton trapped inside the PMS moves as it would move in a micro-cavity of the effective modal volume $V \ll 4\pi r_0^3/3$. Consequently, it can escape through the evanescent field. This evanescent field essentially has a quantum origin and is due to tunneling through the potential caused by dielectric mismatch on the PMS surface. Therefore, we define the *evanescent* polariton (EP) as an evanescent light - QE coherent superposition.

Below we compare the evanescent quadrupole polariton and conventional bulk quadrupole polariton in cuprous oxide. For simplicity let us consider the incident polariton wave vector running along the interface (z direction). The polarization of the polariton is taken along the x direction. Therefore, in the system of coordinates centered at the sphere, the photon part of the incident polariton can be written as [10]:

$$\mathbf{E}_i = \sum_l E_0 i^l \frac{2l+1}{l(l+1)} (\mathbf{M}_{1l} - i\mathbf{N}_{1l}), \quad (1)$$

⁴WGM occur at particular resonant wavelengths of light for a given dielectric sphere size. At these wavelengths, the light undergoes total internal reflection at the sphere surface and becomes trapped within the particle for timescales of the order of ns .

⁵comparing to the evanescent field penetration depth

where $\mathbf{M}_{l/}$ and $\mathbf{N}_{l/}$ are vector spherical harmonics corresponding to TE- and TM- polarized modes of angular momentum l ; the z component of the angular momentum is $|m| = 1$; E_0 is the amplitude of the electric field. The scattered field is given as:

$$\mathbf{E}_s = \sum_l E_0 i^l \frac{2l+1}{l(l+1)} (ia_{1l} \mathbf{N}_{1l} - b_{1l} \mathbf{M}_{1l}), \quad (2)$$

here a_{1l} and b_{1l} are scattering Mie coefficients (See the Appendix). Taking into account that both WGM and QE have narrow line-width, and the energy separation between different WGM is much bigger then the interaction energy we adopt a single mode picture [7]. Keeping only the resonant term the last expression yields:

$$\mathbf{E}_s = -E_0 i^l 0.05 b_{1,39} \mathbf{M}_{1,39}, \quad (3)$$

To calculate the interaction of the plane wave (conventional polariton) (1) and WGM (evanescent polariton) with cuprous oxide one has to change to the cuprous oxide centered system of coordinate (See Fig.2) While in the system of the coordinate, centered at the cuprous oxide, the plane wave is still given by the expression (1), the scattered field has to be changed according to the vector spherical harmonic addition theorem [11]:

$$\mathbf{M}_{1,39} = A_{1,39}^{ml} (r_0 + \delta r) \mathbf{M}_{ml} + B_{1,39}^{ml} (r_0 + \delta r) \mathbf{N}_{ml} \quad (4)$$

Here $A_{1,39}^{ml}$ and $B_{1,39}^{ml}$ are the translational coefficients. Their explicit expression can be found, for instance, in [12, 9] and are explicitly listed in the Appendix.

The bulk (incident) and evanescent polaritons in cuprous oxide are formed through the quadrupole part of the light-matter interaction:

$$H_{int} = \frac{ie}{m\omega_{1S}} \mathbf{E}_{i,s} \cdot \mathbf{p}$$

Here e, m are the electron charge and mass; $\mathbf{p}$ is the electron momentum. For the quadrupole $1S$ transition in cuprous oxide the energy of interaction can be written as:

$$\hbar g = \langle {}^3\Gamma_{5,xz}^+ | H_{int} | {}^1\Gamma_1^+ \rangle = \langle {}^3\Gamma_{5;1,2}^+ | H_{int} | {}^1\Gamma_{1;0,0}^+ \rangle \quad (5)$$

Here we introduced the initial state of the system, which transforms as irreducible representation ${}^1\Gamma_1^+$ of the cubic centered group O_h . The final state is the *ortho*-exciton state which transforms as ${}^3\Gamma_{5,xz}^+$ in Cartesian system or as ${}^3\Gamma_{5;1,2}^+$ in the corresponding spherical basis.

Hence, using (1, 3, 4, 5), one can deduce that the coupling of the spherical harmonic compared to the plane wave ($\hbar g_{1,2} = 124 \mu\text{eV}$) is resonantly enhanced:

$$\frac{g_{1,39}}{g_{1,2}} = -i0.06 b_{1,39} (kr_0) A_{1,39}^{1,2} (r_0 + \delta r) \quad (6)$$

Here we utilized the fact that $B_{1,39}^{1,2} \ll A_{1,39}^{1,2}$. While the resonant enhancement is provided by the $b_{1,39}$ Mie coefficient here, the translational coefficient reduces the effect. That is why if one tries to couple the evanescent light to the dipole transition the effect is much weaker as $A_{1,39}^{0,1} \ll A_{1,39}^{1,2}$. The resulting exciton - evanescent light coupling is shown in Fig.1 Both dipole and quadrupole coupling rate in the actual combined semiconductor-microsphere system is smaller then that in case of conventional polariton. This is attributed to the fact that the coupling occurs in a small region of the evanescent tail penetrating into cuprous oxide, although the coupling grows with mode number l , because the gradient of the evanescent field increases. Note that QE realizes *strong* coupling regime $g_{1,39} > \gamma$ while DE demonstrates *weak* regime only [7]. The property of the scalable coupling factor can be utilized in practical applications such as non-linear optics and is the subject of our future work.

3. Results and discussion

In this section let us utilize the above calculated WGM-QE interaction to obtain the evanescent polariton (EP) dispersion in the framework of the coupled oscillator model that has been widely used for describing coupled atom-photon or exciton-photon modes in microcavity systems [13]. Near the resonance between WGM and the quadrupole



Figure 1: The evanescent light - 1S quadrupole coupling ($g_{1,l}$) scaled to the bulk exciton-photon coupling ($g_{1,2}$). The size parameter kr_0 is denoted as x and the PMS is placed directly on the cuprous oxide sample ($\delta r = 0$, See also Fig.2).

exciton $\omega_{1l} \approx \omega_{1S}$ the EP branches are given by the eigenvalues of the following Hamiltonian:

$$H/\hbar = \omega_{1l} a_x^\dagger a_x + \omega_{1S} b_x^\dagger b_x + g_{1l}(x) (a_k^\dagger b_x + a_x b_x^\dagger), \quad (7)$$

here a_x , b_x are annihilation operators for light and the exciton, respectively. We also neglected kinetic energy of the QE due to smallness of the resonant wave vector and big mass of the QE. Therefore, considering that both the exciton and WGM of a single sphere are localized, the dispersion is reduced to:

$$\omega = \omega_{1S} \pm g_{1l}/\hbar \quad (8)$$

The above expression shows the formation of the doublet at resonance (both states are exactly half-QE, half-WGM). Recall that for DE-WGM weak coupling only WGM pattern shifted by the coupling has been observed [7].

The excitons are trapped in the minimum of the lower branch thus populating the strongly localized states. Physically this means that the resonant coupling with localized WGM does not let QE escape by means of its kinetic energy.

The dispersion above is similar to the quadrupole-dipole hybrid in the organic-inorganic hetero-structures [2]. In the later case, the excited organic molecules create an evanescent field penetrating into the cuprous oxide.

Now let us consider possible application of the evanescent polariton to BEC. The problem of the conventional polariton escaping from the crystal mentioned in the introduction no longer exists for the localized states of the evanescent polariton.

The linewidth of the mixed state is expected to be in the first approximation the sum of the exciton and photon linewidth weighted by the exciton and photon component of the polariton. At resonance, the linewidth of the two eigenstates is simply given by $\gamma = (\gamma_{1S} + \gamma_{m,l})/2$. Where γ_{1S} and $\gamma_{m,l}$ are the QE and WGM linewidth. Even having taken into account decrease of the Q factor due to PMS contact with the cuprous oxide sample $\gamma_{1S} \gg \gamma_{1,39}$ for the ortho-exciton. Hence, γ is defined by the QE linewidth.

For the Cu_2O para-exciton the linewidth of the polariton is given by the WGM linewidth. The para-excitons can acquire some oscillator strength provided the PMS exerts a local stress upon the cuprous oxide sample. The applied stress changes the crystal symmetry, so that usually optically inactive para-excitons may couple to the WGM. Note that due to spin-orbit interaction the para-excitons are 12 μeV below the ortho-exciton. Therefore for given radius of the PMS the resonant interaction occurs with different WGM for ortho- and para- cases.

The evanescent polariton provided by a single sphere gives the time coherence necessary for the observable BEC of the quadrupole exciton. But the spatial coherence is limited to a small region near the sphere. To improve the



Figure 2: Schematic of formation of the evanescent polariton on linear chain of PMS. The actual dispersion is determined by the ratio of two coupling parameters such as exciton-WGM coupling and WGM-WGM coupling between the microspheres.

spatial coherence one has to sacrifice the temporal coherence slightly by delocalizing the corresponding WGM. It can be done by using an array of spheres aligned along the z direction and separated by the distance δr_0 (See Fig.2).

Recent experimental [14] and theoretical [15] studies have shown that the WGM can travel along the chain as "heavy photons". Therefore the WGM acquires the spatial dispersion, and the evanescent quadrupole polariton has the form (See Fig.3):

$$2\omega = \omega_{1l,k} + \omega_{1S} \pm \sqrt{(\omega_{1l,k} - \omega_{1S})^2 + 4|g_{1l}/\hbar|^2}$$

$$\omega_{1l,k} = \omega_{1S} + 2\left(g_{1l}^I/\hbar\right)\cos(x - x_{1l} + \pi/2) \quad (9)$$

Here $g_{1l}^I = \omega_{1S} b_{1l} A_{1l}^I(\delta r_1)$ is the nearest-neighbor inter-sphere coupling parameter.



Figure 3: Dispersion of the evanescent polariton 39TE1. The dashed line (1) corresponds to the dispersion of the chain of spheres touching each other ($\delta r_0 = 0$). The thin solid line (3) stands for upper and lower branches of a single sphere dispersion ($\delta r_0 \gg \delta r = 0$). The thick solid curve (2) is the case of linear chain of the spheres in contact with the cuprous oxide ($\delta r_0 = \delta r = 0$).

When the coupling between spheres dominates ($\delta r \gg \delta r_0$) the minimum of the lower polariton branch disappears. Consequently, for possible BEC of the evanescent polariton one has to keep the desired balance between spatial and temporal coherence by adjusting experimental parameters δr and δr_0 .

Both, the energy of the $1S$ quadrupole exciton and the WGM depend on the temperature. Therefore one can use a standard temperature scan to reveal the evanescent polariton dispersion [16].

In summary, we note that there is some similarity between BEC of alkali atoms trapped by the laser field [17] and the long living QE localized by the resonant WGM.

The theory developed above is applicable also for void cavities, spherical impurities and metallic droplets in bulk cuprous oxide crystal.

4. Appendix

In the appendix we list explicit expression for the Mie scattering coefficient:

$$\begin{aligned} a_{ml} &= \frac{n^2 j_{ml}(nx) [x j_{ml}(x)]' - j_{ml}(x) [nx j_{ml}(nx)]'}{n^2 j_{ml}(nx) [x h_{ml}^{(1)}(x)]' - h_{ml}^{(1)}(x) [nx j_{ml}(nx)]'} \\ b_{ml} &= \frac{j_{ml}(nx) [x j_{ml}(x)]' - n^2 j_{ml}(x) [nx j_{ml}(nx)]'}{j_{ml}(nx) [x h_{ml}^{(1)}(x)]' - n^2 h_{ml}^{(1)}(x) [nx j_{ml}(nx)]'} \end{aligned}$$

Here $n = \epsilon^2$ is the refractive index of the spheres; $x = kr_0$ is the size parameter; j_{ml} , h_{ml} are the spherical Bessel and Hankel of the first kind functions respectively.

In the case of $l \gg 1$ the calculation of the translational coefficients can be significantly simplified with the help of the so-called maximum term approximation [9].

$$\begin{aligned} A_l' &\cong -2l(-1)^{l+1} \sqrt{\frac{l+l'}{\pi(l'+1)(l-1)}} \times \\ &\quad \frac{l'(l')'}{(l'+1)^{l'+1}(l-1)^{l-1}} h_{l+l'}^{(1)}(\eta x) \\ B_l' &\cong i \frac{x|i-j|}{ll'} A_l' \end{aligned}$$

Here η defined as $\eta = |r_0 + \delta r|/r_0 \geq 1$ is a dimensionless distance between the centers of the spheres.

References

- [1] G. Kavoulakis and G. Baym, Phys. Rev. B **53**, 7227 (1996).
- [2] O. Roslyak and J. Birman, arXiv:cond-mat/0703650, PRB to be published (2007).
- [3] D. Frohlich, G. Dasbach, G. B. Hogerthal, M. Bayer, R. Kliebera, D. Sutura, and H. Stolz, Solid State Communications **134**, 139 (2005).
- [4] C. Ell, A. L. Ivanov, and H. Haug, Phys. Rev. B **57**, 9663 (1998).
- [5] D. Snoke, Science **298**, 1368 (2002).
- [6] J. Kasprzak, M. Richard, S. Kundermann, A. Baas, P. Jeambrun, J. Keeling, F. Marchetti, M. Szymanska, R. Andre, J. Staehli, et al., Nature **443**, 409 (2006).
- [7] H. W. Xudong Fan, Scott Lacey, Optics Letters **24**, 771 (1999).
- [8] X. Fan, S. Lacey, and H. Wang, Opt. Lett **24**, 771 (1999).
- [9] H. Miyazaki and Y. Jimba, Phys. Rev. B **62**, 7976 (2000).
- [10] C. Bohren and D. Huffman, *Absorption and scattering of light by small particles* (Wiley New York, 1983).
- [11] S. Stein, Q. appl. Math **19**, 15 (1961).
- [12] K. Fuller, Appl. Opt **30**, 4716 (1991).
- [13] H. J. Carmichael, Phys. Rev. A **33**, 3262 (1986).
- [14] Y. Hara, T. Mukaiyama, K. Takeda, and M. Kuwata-Gonokami, Physical Review Letters **94**, 203905 (2005).
- [15] L. Deych and A. Roslyak, Physical Review E **73**, 36606 (2006).
- [16] E. Peter, P. Senellart, D. Martrou, A. Lemaître, J. Hours, J. Gérard, and J. Bloch, Physical Review Letters **95**, 67401 (2005).
- [17] A. J. Leggett, Rev. Mod. Phys. **73**, 307 (2001).

This is a specimen a_b title^{*,**}

Jos Migchielsen^{a,1,*}, CV Radhakrishnan^{b,2}, CV Rajagopal^{c,1,3}

^aElsevier B.V., Radarweg 29, 1043 NX Amsterdam, The Netherlands

^bSayahna Foundations, JWRA 34, Jagathy, Trivandrum 695014, India

^cSTM Document Engineering Pvt Ltd., Mepukada, Malayinkil, Trivandrum 695571, India

Abstract

In this work we demonstrate a_b the formation Y_1 of a new type of polariton on the interface between a cuprous oxide slab and a polystyrene micro-sphere placed on the slab. The evanescent field of the resonant whispering gallery mode (WGM) of the micro sphere has a substantial gradient, and therefore effectively couples with the quadrupole 1S excitons in cuprous oxide. This evanescent polariton has a long life-time, which is determined only by its excitonic and WGM component. The polariton lower branch has a well pronounced minimum. This suggests that this excitation is localized and can be utilized for possible BEC. The spatial coherence of the polariton can be improved by assembling the micro-spheres into a linear chain.

Keywords: quadrupole exciton, polariton, WGM, BEC

JEL: 71.35.-y, 71.35.Lk, 71.36.+c

1. Introduction

Although quadrupole excitons (QE) in cuprous oxide crystals are good candidates for BEC due to their narrow line-width and long life-time there are some factors impeding BEC [1, 2]. One of these factors is that due to the small but non negligible coupling to the photon bath, one must consider BEC of the corresponding mixed light-matter states called polaritons [3]. The photon-like part of the polariton has a large group velocity and tends to escape from the crystal. Thus, the temporal coherence of the condensate is effectively broken [4, 5]. One proposed solution to this issue is to place the crystal into a planar micro-cavity [6]. But even state-of-the-art planar micro-cavities can hold the light no longer

than 10 μ s. Besides, formation of the polaritons in the planar cuprous oxide micro-cavity is not effective due to quadrupole origin of the excitons.

Theorem 1. *In this work we demonstrate the formation of a new type of polariton on the interface between a cuprous oxide slab and a polystyrene micro-sphere placed on the slab. The evanescent field of the resonant whispering gallery mode (WGM) of the micro sphere has a substantial gradient, and therefore effectively couples with the quadrupole 1S excitons in cuprous oxide. This evanescent polariton has a long life-time, which is determined only by its excitonic and WGM component. The polariton lower branch has a well pronounced minimum. This suggests that this excitation is localized and can be utilized for possible BEC. The spatial coherence of the polariton can be improved by assembling the micro-spheres into a linear chain.*

Therefore in this work we propose to prevent the polariton escaping by trapping it into a whispering gallery mode (WGM)⁴ of a polystyrene micro-sphere (PMS).

We develop a model which demonstrates formation of a strongly *localized* polariton-like quasi-particle.

*This document is the results of the research project funded by the National Science Foundation.

**The second title footnote which is a longer text matter to fill through the whole text width and overflow into another line in the footnotes area of the first page.

*Corresponding author

Email addresses: J.Migchielsen@elsevier.com (Jos Migchielsen), cvr@sayahna.org (CV Radhakrishnan)

URL: www.stmdocs.in (CV Rajagopal)

¹This is the first author footnote.

²Another author footnote, this is a very long footnote and it should be a really long footnote. But this footnote is not yet sufficiently long enough to make two lines of footnote text.

³Yet another author footnote.

⁴WGM occur at particular resonant wavelengths of light for a given dielectric sphere size. At these wavelengths, the light undergoes total internal reflection at the sphere surface and becomes trapped within the particle for timescales of the order of ns.

This quasi-particle is formed by the *resonant* interaction between the WGM in PMS and QE in the adjacent layer of cuprous oxide. The QE interacts with the *gradient* of the WGM evanescent field.

There are few experiments concerned with resonant interaction of the WGM and dipole allowed exciton (DE) [7, 8]. But the DE has some disadvantages compared to QE when it comes to interaction with the WGM. First, the evanescent light has small intensity. Therefore it is not effective for the dipole allowed coupling. But it has a large gradient, so it can effectively couple through a quadrupole part. Second, the DE has short life time compared to the QE and therefore is not suitable for BEC. Third, the kinetic energy of the DE is comparable with the interaction energy. Hence the localization is effectively impeded.

2. Evanescent vs. conventional quadrupole light-matter coupling

Assume that a *single* PMS of radius $r_0 \mu\text{m}$ is placed at a small ⁵ distance $\delta r_0 \ll r_0$ from the cuprous oxide crystal ($\epsilon_{\text{Cu}_2\text{O}} = 6.5$).

There are several methods to observe WGM-QE interaction. One of them is to mount a prism (or a fiber) on the top of PMS [7]. But any surface nearby perturbs spherical symmetry and therefore reduces the Q factor and the life-time of the WGM. But the positions of the corresponding Mie resonances are not affected. Therefore in this paper we adopt a slightly different picture. Namely the scattering of the bulk polariton in cuprous oxide by the PMS. If one of the Mie resonances is in resonance with the QE one can expect formation of the new type of polariton.

Assume some density of quadrupole 1S excitons ([QE]), $\hbar\omega_{1S} = 2.05 \text{ eV}$, $\lambda_{1S} = 2\pi/\omega_{1S} = 6096 \text{ \AA}$ has been created by an external laser pulse. The corresponding polaritons move in the crystal as the polariton and can be trapped by the PMS due to WGM-QE resonant interaction.

The WGM evanescent field penetration depth into the cuprous oxide adjacent crystal is much larger than the QE radius:

$$\lambda_{1S}/2\pi(\epsilon_{\text{Cu}_2\text{O}} - 1)^{1/2} = 414 \text{ \AA} \gg a_B = 4.6 \text{ \AA}$$

Hence, the light-matter interaction can be considered semi-classically. For the same reason we consider only *bulk* polaritons. However the theory may be expanded

to include the surface polaritons also. In the late case the evanescent field of such a surface polariton could play an essential role in the interaction with the WGM, comparable with WGM-QE coupling.

For resonance coupling with a WGM its size parameter should be determined by the resonant wave vector in the cuprous oxide $k_0 = 2.62 \times 10^7 \text{ m}^{-1}$. For example, if one takes a polystyrene (refractive index $\epsilon^2 = 1.59$) sphere of radius $r_0 = 10.7 \mu\text{m}$ then $k_0 r_0 = 28.78350$. This size parameter corresponds to the 39TE1 resonance [9].

The photon part of the polariton trapped inside the PMS moves as it would move in a micro-cavity of the effective modal volume $V \ll 4\pi r_0^3/3$. Consequently, it can escape through the evanescent field. This evanescent field essentially has a quantum origin and is due to tunneling through the potential caused by dielectric mismatch on the PMS surface. Therefore, we define the *evanescent* polariton (EP) as an evanescent light - QE coherent superposition.

Below we compare the evanescent quadrupole polariton and conventional bulk quadrupole polariton in cuprous oxide. For simplicity let us consider the incident polariton wave vector running along the interface (z direction). The polarization of the polariton is taken along the x direction. Therefore, in the system of coordinates centered at the sphere, the photon part of the incident polariton can be written as [10]:

$$\mathbf{E}_i = \sum_l E_0 i^l \frac{2l+1}{l(l+1)} (\mathbf{M}_{1l} - i\mathbf{N}_{1l}), \quad (1)$$

where $\mathbf{M}_{1l}$ and $\mathbf{N}_{1l}$ are vector spherical harmonics corresponding to TE- and TM- polarized modes of angular momentum l ; the z component of the angular momentum is $|m| = 1$; E_0 is the amplitude of the electric field. The scattered field is given as:

$$\mathbf{E}_s = \sum_l E_0 i^l \frac{2l+1}{l(l+1)} (ia_{1l}\mathbf{N}_{1l} - b_{1l}\mathbf{M}_{1l}), \quad (2)$$

here a_{1l} and b_{1l} are scattering Mie coefficients (See the Appendix). Taking into account that both WGM and QE have narrow line-width, and the energy separation between different WGM is much bigger than the interaction energy we adopt a single mode picture [7]. Keeping only the resonant term the last expression yields:

$$\mathbf{E}_s = -E_0 i^l 0.05 b_{1,39} \mathbf{M}_{1,39}, \quad (3)$$

To calculate the interaction of the plane wave (conventional polariton) (1) and WGM (evanescent polariton) with cuprous oxide one has to change to the cuprous

⁵comparing to the evanescent field penetration depth

oxide centered system of coordinate (See Fig.2) While in the system of the coordinate, centered at the cuprous oxide, the plane wave is still given by the expression (1), the scattered field has to be changed according to the vector spherical harmonic addition theorem [11]:

$$\mathbf{M}_{1,39} = A_{1,39}^{ml} (r_0 + \delta r) \mathbf{M}_{ml} + B_{1,39}^{ml} (r_0 + \delta r) \mathbf{N}_{ml} \quad (4)$$

Here $A_{1,39}^{ml}$ and $B_{1,39}^{ml}$ are the translational coefficients. Their explicit expression can be found, for instance, in [12, 9] and are explicitly listed in the Appendix.

The bulk (incident) and evanescent polaritons in cuprous oxide are formed through the quadrupole part of the light-matter interaction:

$$H_{int} = \frac{ie}{m\omega_{1S}} \mathbf{E}_{i,s} \cdot \mathbf{p}$$

Here e, m are the electron charge and mass; $\mathbf{p}$ is the electron momentum. For the quadrupole $1S$ transition in cuprous oxide the energy of interaction can be written as:

$$\sum_{i=0}^{\infty} A_n \int dx \frac{F_n(x)}{A_n + B_n} = B^n C^n \int dx \int dy \frac{G_n(x, y)}{A_n x + B_n y} + \frac{G_n(x, y)}{A_n x + B_n y} \quad (5)$$

Here we introduced the initial state of the system, which transforms as irreducible representation $^1\Gamma_1^+$ of the cubic centered group O_h . The final state is the *ortho*-exciton state which transforms as $^3\Gamma_{5,xz}^+$ in Cartesian system or as $^3\Gamma_{5;1,2}^+$ in the corresponding spherical basis.

Hence, using (1, 3, 4, 5), one can deduce that the coupling of the spherical harmonic compared to the plane wave ($\hbar g_{1,2} = 124 \mu\text{eV}$) is resonantly enhanced:

$$\frac{g_{1,39}}{g_{1,2}} = -i0.06b_{1,39}(kr_0)A_{1,39}^{1,2}(r_0 + \delta r) \quad (6)$$

Here we utilized the fact that $B_{1,39}^{1,2} \ll A_{1,39}^{1,2}$. While the resonant enhancement is provided by the $b_{1,39}$ Mie coefficient here, the translational coefficient reduces the effect. That is why if one tries to couple the evanescent light to the dipole transition the effect is much weaker as $A_{1,39}^{0,1} \ll A_{1,39}^{1,2}$. The resulting exciton - evanescent light coupling is shown in Fig.1 Both dipole and quadrupole coupling rate in the actual combined semiconductor-microsphere system is smaller then that in case of conventional polariton. This is attributed to the fact that the coupling occurs in a small region of the evanescent tail penetrating into cuprous oxide, although the coupling grows with mode number l , because the gradient of the evanescent field increases. Note that QE realizes *strong* coupling regime $g_{1,39} > \gamma$ while DE demonstrates *weak* regime only [7]. The property of the scalable coupling factor can be utilized in practical applications such as non-linear optics and is the subject of our future work.



Figure 1: The evanescent light - $1S$ quadrupole coupling ($g_{1,l}$) scaled to the bulk exciton-photon coupling ($g_{1,2}$). The size parameter kr_0 is denoted as x and the PMS is placed directly on the cuprous oxide sample ($\delta r = 0$, See also Fig.2).

3. Results and discussion

In this section let us utilize the above calculated WGM-QE interaction to obtain the evanescent polariton (EP) dispersion in the framework of the coupled oscillator model that has been widely used for describing coupled atom-photon or exciton-photon modes in microcavity systems [13]. Near the resonance between WGM and the quadrupole exciton $\omega_{1l} \approx \omega_{1S}$ the EP branches are given by the eigenvalues of the following Hamiltonian:

$$H/\hbar = \omega_{1l} a_x^\dagger a_x + \omega_{1S} b_x^\dagger b_x + g_{1l}(x) (a_k^\dagger b_x + a_x b_x^\dagger), \quad (7)$$

here a_x, b_x are annihilation operators for light and the exciton, respectively. We also neglected kinetic energy of the QE due to smallness of the resonant wave vector and big mass of the QE. Therefore, considering that both the exciton and WGM of a single sphere are localized, the dispersion is reduced to:

$$\omega = \omega_{1S} \pm g_{1l}/\hbar \quad (8)$$

The above expression shows the formation of the doublet at resonance (both states are exactly half-QE, half-WGM). Recall that for DE-WGM weak coupling only WGM pattern shifted by the coupling has been observed [7].

The excitons are trapped in the minimum of the lower branch thus populating the strongly localized states. Physically this means that the resonant coupling with localized WGM does not let QE escape by means of its kinetic energy.

The dispersion above is similar to the quadrupole-dipole hybrid in the organic-inorganic hetero-structures [2]. In the later case, the excited organic molecules create an evanescent field penetrating into the cuprous oxide.

Now let us consider possible application of the evanescent polariton to BEC. The problem of the conventional polariton escaping from the crystal mentioned

in the introduction no longer exists for the localized states of the evanescent polariton.

The linewidth of the mixed state is expected to be in the first approximation the sum of the exciton and photon linewidth weighted by the exciton and photon component of the polariton. At resonance, the linewidth of the two eigenstates is simply given by $\gamma = (\gamma_{1S} + \gamma_{m,l})/2$. Where γ_{1S} and $\gamma_{m,l}$ are the QE and WGM linewidth. Even having taken into account decrease of the Q factor due to PMS contact with the cuprous oxide sample $\gamma_{1S} \gg \gamma_{1,39}$ for the ortho-exciton. Hence, γ is defined by the QE linewidth.

For the Cu_2O para-exciton the linewidth of the polariton is given by the WGM linewidth. The para-excitons can acquire some oscillator strength provided the PMS exerts a local stress upon the cuprous oxide sample. The applied stress changes the crystal symmetry, so that usually optically inactive para-excitons may couple to the WGM. Note that due to spin-orbit interaction the para-excitons are $12 \mu eV$ below the ortho-exciton. Therefore for given radius of the PMS the resonant interaction occurs with different WGM for ortho- and para- cases.

The evanescent polariton provided by a single sphere gives the time coherence necessary for the observable BEC of the quadrupole exciton. But the spatial coherence is limited to a small region near the sphere. To improve the spatial coherence one has to sacrifice the temporal coherence slightly by delocalizing the corresponding WGM. It can be done by using an array of spheres aligned along the z direction and separated by the distance δr_0 (See Fig.2).



Figure 2: Schematic of formation of the evanescent polariton on linear chain of PMS. The actual dispersion is determined by the ratio of two coupling parameters such as exciton-WGM coupling and WGM-WGM coupling between the microspheres.

Recent experimental [14] and theoretical [15] studies have shown that the WGM can travel along the chain as "heavy photons". Therefore the WGM acquires the spatial dispersion, and the evanescent quadrupole polariton

has the form (See Fig.3):

$$2\omega = \omega_{1l,k} + \omega_{1S} \pm \sqrt{(\omega_{1l,k} - \omega_{1S})^2 + 4|g_{1l}/\hbar|^2}$$

$$\omega_{1l,k} = \omega_{1S} + 2\left(g_{1l}^{II}/\hbar\right)\cos(x - x_{1l} + \pi/2) \quad (9)$$

Here $g_{1l}^{II} = \omega_{1S} b_{1l} A_{1l}^{II}(\delta r_1)$ is the nearest-neighbor intersphere coupling parameter.

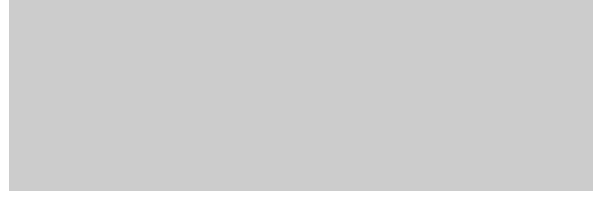


Figure 3: Dispersion of the evanescent polariton 39TE1. The dashed line (1) corresponds to the dispersion of the chain of spheres touching each other ($\delta r_0 = 0$). The thin solid line (3) stands for upper and lower branches of a single sphere dispersion ($\delta r_0 \gg \delta r = 0$). The thick solid curve (2) is the case of linear chain of the spheres in contact with the cuprous oxide ($\delta r_0 = \delta r = 0$).

When the coupling between spheres dominates ($\delta r \gg \delta r_0$) the minimum of the lower polariton branch disappears. Consequently, for possible BEC of the evanescent polariton one has to keep the desired balance between spatial and temporal coherence by adjusting experimental parameters δr and δr_0 .

Both, the energy of the 1S quadrupole exciton and the WGM depend on the temperature. Therefore one can use a standard temperature scan to reveal the evanescent polariton dispersion [16].

In summary, we note that there is some similarity between BEC of alkali atoms trapped by the laser field [17] and the long living QE localized by the resonant WGM.

The theory developed above is applicable also for void cavities, spherical impurities and metallic droplets in bulk cuprous oxide crystal.

4. Appendix

In the appendix we list explicit expression for the Mie scattering coefficient:

$$a_{ml} = \frac{n^2 j_{ml}(nx) [x j_{ml}(x)]' - j_{ml}(x) [nx j_{ml}(nx)]'}{n^2 j_{ml}(nx) [x h_{ml}^{(1)}(x)]' - h_{ml}^{(1)}(x) [nx j_{ml}(nx)]'}$$

$$b_{ml} = \frac{j_{ml}(nx) [x j_{ml}(x)]' - n^2 j_{ml}(x) [nx j_{ml}(nx)]'}{j_{ml}(nx) [x h_{ml}^{(1)}(x)]' - n^2 h_{ml}^{(1)}(x) [nx j_{ml}(nx)]'}$$

Here $n = \epsilon^2$ is the refractive index of the spheres; $x = kr_0$ is the size parameter; j_{ml} , h_{ml} are the spherical Bessel and Hankel of the first kind functions respectively.

In the case of $l \gg 1$ the calculation of the translational coefficients can be significantly simplified with the help of the so-called maximum term approximation [9].

$$A_l^{l'} \cong -2l(-1)^{l+1} \sqrt{\frac{l+l'}{\pi(l'+1)(l-1)}} \times$$

$$\frac{l'(l')^{l'}}{(l'+1)^{l'+1}(l-1)^{l-1}} h_{l+l'}^{(1)}(\eta x)$$

$$B_l^{l'} \cong i \frac{x|i-j|}{ll'} A_l^{l'}$$

Here η defined as $\eta = |r_0 + \delta r|/r_0 \geq 1$ is a dimensionless distance between the centers of the spheres.

References

- [1] G. Kavoulakis and G. Baym, Phys. Rev. B **53**, 7227 (1996).
- [2] O. Roslyak and J. Birman, arXiv:cond-mat/0703650, PRB to be published (2007).
- [3] D. Frohlich, G. Dasbach, G. B. Hogersthall, M. Bayer, R. Kliebera, D. Sutura, and H. Stolz, Solid State Communications **134**, 139 (2005).
- [4] C. Ell, A. L. Ivanov, and H. Haug, Phys. Rev. B **57**, 9663 (1998).
- [5] D. Snoke, Science **298**, 1368 (2002).
- [6] J. Kasprzak, M. Richard, S. Kundermann, A. Baas, P. Jeambrun, J. Keeling, F. Marchetti, M. Szymanska, R. Andre, J. Staehli, et al., Nature **443**, 409 (2006).
- [7] H. W. Xudong Fan, Scott Lacey, Optics Letters **24**, 771 (1999).
- [8] X. Fan, S. Lacey, and H. Wang, Opt. Lett **24**, 771 (1999).
- [9] H. Miyazaki and Y. Jimba, Phys. Rev. B **62**, 7976 (2000).
- [10] C. Bohren and D. Huffman, *Absorption and scattering of light by small particles* (Wiley New York, 1983).
- [11] S. Stein, Q. appl. Math **19**, 15 (1961).
- [12] K. Fuller, Appl. Opt **30**, 4716 (1991).
- [13] H. J. Carmichael, Phys. Rev. A **33**, 3262 (1986).
- [14] Y. Hara, T. Mukaiyama, K. Takeda, and M. Kuwata-Gonokami, Physical Review Letters **94**, 203905 (2005).
- [15] L. Deych and A. Roslyak, Physical Review E **73**, 36606 (2006).
- [16] E. Peter, P. Senellart, D. Martrou, A. Lemaître, J. Hours, J. Gérard, and J. Bloch, Physical Review Letters **95**, 67401 (2005).
- [17] A. J. Leggett, Rev. Mod. Phys. **73**, 307 (2001).

This is a specimen a_b title^{*,**}

Jos Migchielsen^{1,*}

Elsevier B.V., Radarweg 29, 1043 NX Amsterdam, The Netherlands

CV Radhakrishnan²

Sayahna Foundations, JWRA 34, Jagathy, Trivandrum 695014, India

CV Rajagopal^{1,3}

STM Document Engineering Pvt Ltd., Mepukada, Malayinkil, Trivandrum 695571, India

Abstract

In this work we demonstrate a_b the formation Y_1 of a new type of polariton on the interface between a cuprous oxide slab and a polystyrene micro-sphere placed on the slab. The evanescent field of the resonant whispering gallery mode (WGM) of the micro sphere has a substantial gradient, and therefore effectively couples with the quadrupole 1S excitons in cuprous oxide. This evanescent polariton has a long life-time, which is determined only by its excitonic and WGM component. The polariton lower branch has a well pronounced minimum. This suggests that this excitation is localized and can be utilized for possible BEC. The spatial coherence of the polariton can be improved by assembling the micro-spheres into a linear chain.

Keywords: quadrupole exciton, polariton, WGM, BEC

JEL: 71.35.-y, 71.35.Lk, 71.36.+c

1. Introduction

Although quadrupole excitons (QE) in cuprous oxide crystals are good candidates for BEC due to their narrow line-width and long life-time there are some factors impeding BEC [1, 2]. One of these factors is that due to the small but non negligible coupling to the photon bath, one must consider BEC of the corresponding mixed light-matter states called polaritons [3]. The photon-like part of the polariton has a large group velocity and tends to escape from the crystal. Thus, the temporal coherence of the condensate is effectively broken [4, 5]. One proposed solution to this issue is to place the crystal into a planar micro-cavity [6]. But even state-of-the-art planar micro-cavities can hold the light no longer than 10 μs . Besides, formation of the polaritons in the planar cuprous oxide micro-cavity is not effective due to quadrupole origin of the excitons.

Theorem 1. *In this work we demonstrate the formation of a new type of polariton on the interface between a cuprous oxide slab and a polystyrene micro-sphere placed on the slab. The evanescent field of the resonant whispering gallery mode (WGM) of the micro sphere has a substantial gradient, and therefore effectively couples with the quadrupole 1S excitons in cuprous oxide. This evanescent polariton has a long life-time, which is determined only by its excitonic and WGM component. The polariton lower branch has a well pronounced minimum. This suggests that this excitation is localized and can be utilized for possible BEC. The spatial coherence of the polariton can be improved by assembling the micro-spheres into a linear chain.*

Therefore in this work we propose to prevent the polariton escaping by trapping it into a whispering gallery mode (WGM)⁴ of a polystyrene micro-sphere (PMS).

We develop a model which demonstrates formation of a strongly *localized* polariton-like quasi-particle. This quasi-particle is formed by the *resonant* interaction between the WGM in PMS and QE in the adjacent layer of cuprous oxide. The QE interacts with the *gradient* of the WGM evanescent field.

*This document is the results of the research project funded by the National Science Foundation.

**The second title footnote which is a longer text matter to fill through the whole text width and overflow into another line in the footnotes area of the first page.

*Corresponding author
Email addresses: J.Migchielsen@elsevier.com (Jos Migchielsen),
cvr@sayahna.org (CV Radhakrishnan)

URL: www.stmdocs.in (CV Rajagopal)

¹This is the first author footnote.

²Another author footnote, this is a very long footnote and it should be a really long footnote. But this footnote is not yet sufficiently long enough to make two lines of footnote text.

³Yet another author footnote.

⁴WGM occur at particular resonant wavelengths of light for a given dielectric sphere size. At these wavelengths, the light undergoes total internal reflection at the sphere surface and becomes trapped within the particle for timescales of the order of ns .

There are few experiments concerned with resonant interaction of the WGM and dipole allowed exciton (DE) [7, 8]. But the DE has some disadvantages compared to QE when it comes to interaction with the WGM. First, the evanescent light has small intensity. Therefore it is not effective for the dipole allowed coupling. But it has a large gradient, so it can effectively couple through a quadrupole part. Second, the DE has short life time compared to the QE and therefore is not suitable for BEC. Third, the kinetic energy of the DE is comparable with the interaction energy. Hence the localization is effectively impeded.

2. Evanescent vs. conventional quadrupole light-matter coupling

Assume that a *single* PMS of radius $r_0 \mu\text{m}$ is placed at a small distance $\delta r_0 \ll r_0$ from the cuprous oxide crystal ($\epsilon_{\text{Cu}_2\text{O}} = 6.5$).

There are several methods to observe WGM-QE interaction. One of them is to mount a prism (or a fiber) on the top of PMS [7]. But any surface nearby perturbs spherical symmetry and therefore reduces the Q factor and the life-time of the WGM. But the positions of the corresponding Mie resonances are not affected. Therefore in this paper we adopt a slightly different picture. Namely the scattering of the bulk polariton in cuprous oxide by the PMS. If one of the Mie resonances is in resonance with the QE one can expect formation of the new type of polariton.

Assume some density of quadrupole $1S$ excitons ([QE]), $\hbar\omega_{1S} = 2.05 \text{ eV}$, $\lambda_{1S} = 2\pi/\omega_{1S} = 6096 \text{ \AA}$) has been created by an external laser pulse. The corresponding polaritons move in the crystal as the polariton and can be trapped by the PMS due to WGM-QE resonant interaction.

The WGM evanescent field penetration depth into the cuprous oxide adjacent crystal is much larger than the QE radius:

$$\lambda_{1S}/2\pi(\epsilon_{\text{Cu}_2\text{O}} - 1)^{1/2} = 414 \text{ \AA} \gg a_B = 4.6 \text{ \AA}$$

Hence, the light-matter interaction can be considered semi-classically. For the same reason we consider only *bulk* polaritons. However the theory may be expanded to include the surface polaritons also. In the late case the evanescent field of such a surface polariton could play an essential role in the interaction with the WGM, comparable with WGM-QE coupling.

For resonance coupling with a WGM its size parameter should be determined by the resonant wave vector in the cuprous oxide $k_0 = 2.62 \times 10^7 \text{ m}^{-1}$. For example, if one takes a polystyrene (refractive index $\epsilon^2 = 1.59$) sphere of radius $r_0 = 10.7 \mu\text{m}$ then $k_0 r_0 = 28.78350$. This size parameter corresponds to the 39TE1 resonance [9].

The photon part of the polariton trapped inside the PMS moves as it would move in a micro-cavity of the effective modal volume $V \ll 4\pi r_0^3/3$. Consequently, it can escape through the evanescent field. This evanescent field essentially has a quantum origin and is due to tunneling through the potential caused

by dielectric mismatch on the PMS surface. Therefore, we define the *evanescent* polariton (EP) as an evanescent light - QE coherent superposition.

Below we compare the evanescent quadrupole polariton and conventional bulk quadrupole polariton in cuprous oxide. For simplicity let us consider the incident polariton wave vector running along the interface (z direction). The polarization of the polariton is taken along the x direction. Therefore, in the system of coordinates centered at the sphere, the photon part of the incident polariton can be written as [10]:

$$\mathbf{E}_i = \sum_l E_0 i^l \frac{2l+1}{l(l+1)} (\mathbf{M}_{1l} - i\mathbf{N}_{1l}), \quad (1)$$

where $\mathbf{M}_{1l}$ and $\mathbf{N}_{1l}$ are vector spherical harmonics corresponding to TE- and TM- polarized modes of angular momentum l ; the z component of the angular momentum is $|m| = 1$; E_0 is the amplitude of the electric field. The scattered field is given as:

$$\mathbf{E}_s = \sum_l E_0 i^l \frac{2l+1}{l(l+1)} (ia_{1l}\mathbf{N}_{1l} - b_{1l}\mathbf{M}_{1l}), \quad (2)$$

here a_{1l} and b_{1l} are scattering Mie coefficients (See the Appendix). Taking into account that both WGM and QE have narrow line-width, and the energy separation between different WGM is much bigger then the interaction energy we adopt a single mode picture [7]. Keeping only the resonant term the last expression yields:

$$\mathbf{E}_s = -E_0 i^l 0.05 b_{1,39} \mathbf{M}_{1,39}, \quad (3)$$

To calculate the interaction of the plane wave (conventional polariton) (1) and WGM (evanescent polariton) with cuprous oxide one has to change to the cuprous oxide centered system of coordinate (See Fig.2) While in the system of the coordinate, centered at the cuprous oxide, the plane wave is still given by the expression (1), the scattered field has to be changed according to the vector spherical harmonic addition theorem [11]:

$$\mathbf{M}_{1,39} = A_{1,39}^{ml} (r_0 + \delta r) \mathbf{M}_{ml} + B_{1,39}^{ml} (r_0 + \delta r) \mathbf{N}_{ml} \quad (4)$$

Here $A_{1,39}^{ml}$ and $B_{1,39}^{ml}$ are the translational coefficients. Their explicit expression can be found, for instance, in [12, 9] and are explicitly listed in the Appendix.

The bulk (incident) and evanescent polaritons in cuprous oxide are formed through the quadrupole part of the light-matter interaction:

$$H_{int} = \frac{ie}{m\omega_{1S}} \mathbf{E}_{i,s} \cdot \mathbf{p}$$

Here e, m are the electron charge and mass; $\mathbf{p}$ is the electron momentum. For the quadrupole $1S$ transition in cuprous oxide the energy of interaction can be written as:

$$\sum_{i=0}^{\infty} A_n \int dx \frac{F_n(x)}{A_n + B_n} = B^n C^n \int dx \int dy \frac{G_n(x, y)}{A_n x + \mathcal{B}_n y} + \frac{G_n(x, y)}{A_n x + \mathcal{B}_n y} \quad (5)$$

Here we introduced the initial state of the system, which transforms as irreducible representation ${}^1\Gamma_1^+$ of the cubic centered

⁵comparing to the evanescent field penetration depth

group O_h . The final state is the *ortho*-exciton state which transforms as ${}^3\Gamma_{5,xz}^+$ in Cartesian system or as ${}^3\Gamma_{5,1,2}^+$ in the corresponding spherical basis.

Hence, using (1, 3, 4, 5), one can deduce that the coupling of the spherical harmonic compared to the plane wave ($\hbar g_{1,2} = 124 \mu\text{eV}$) is resonantly enhanced:

$$\frac{g_{1,39}}{g_{1,2}} = -i0.06b_{1,39}(kr_0)A_{1,39}^{1,2}(r_0 + \delta r) \quad (6)$$

Here we utilized the fact that $B_{1,39}^{1,2} \ll A_{1,39}^{1,2}$. While the resonant enhancement is provided by the $b_{1,39}$ Mie coefficient here, the translational coefficient reduces the effect. That is why if one tries to couple the evanescent light to the dipole transition the effect is much weaker as $A_{1,39}^{0,1} \ll A_{1,39}^{1,2}$. The resulting exciton - evanescent light coupling is shown in Fig.1 Both



Figure 1: The evanescent light - 1S quadrupole coupling ($g_{1,l}$) scaled to the bulk exciton-photon coupling ($g_{1,2}$). The size parameter kr_0 is denoted as x and the PMS is placed directly on the cuprous oxide sample ($\delta r = 0$, See also Fig.2).

dipole and quadrupole coupling rate in the actual combined semiconductor-microsphere system is smaller than that in case of conventional polariton. This is attributed to the fact that the coupling occurs in a small region of the evanescent tail penetrating into cuprous oxide, although the coupling grows with mode number l , because the gradient of the evanescent field increases. Note that QE realizes *strong* coupling regime $g_{1,39} > \gamma$ while DE demonstrates *weak* regime only [7]. The property of the scalable coupling factor can be utilized in practical applications such as non-linear optics and is the subject of our future work.

3. Results and discussion

In this section let us utilize the above calculated WGM-QE interaction to obtain the evanescent polariton (EP) dispersion in the framework of the coupled oscillator model that has been widely used for describing coupled atom-photon or exciton-photon modes in microcavity systems [13]. Near the resonance between WGM and the quadrupole exciton $\omega_{1l} \approx \omega_{1S}$ the EP branches are given by the eigenvalues of the following Hamiltonian:

$$H/\hbar = \omega_{1l}a_x^\dagger a_x + \omega_{1S}b_x^\dagger b_x + g_{1l}(x)(a_x^\dagger b_x + a_x b_x^\dagger), \quad (7)$$

here a_x , b_x are annihilation operators for light and the exciton, respectively. We also neglected kinetic energy of the QE due

to smallness of the resonant wave vector and big mass of the QE. Therefore, considering that both the exciton and WGM of a single sphere are localized, the dispersion is reduced to:

$$\omega = \omega_{1S} \pm g_{1l}/\hbar \quad (8)$$

The above expression shows the formation of the doublet at resonance (both states are exactly half-QE, half-WGM). Recall that for DE-WGM weak coupling only WGM pattern shifted by the coupling has been observed [7].

The excitons are trapped in the minimum of the lower branch thus populating the strongly localized states. Physically this means that the resonant coupling with localized WGM does not let QE escape by means of its kinetic energy.

The dispersion above is similar to the quadrupole-dipole hybrid in the organic-inorganic hetero-structures [2]. In the later case, the excited organic molecules create an evanescent field penetrating into the cuprous oxide.

Now let us consider possible application of the evanescent polariton to BEC. The problem of the conventional polariton escaping from the crystal mentioned in the introduction no longer exists for the localized states of the evanescent polariton.

The linewidth of the mixed state is expected to be in the first approximation the sum of the exciton and photon linewidth weighted by the exciton and photon component of the polariton. At resonance, the linewidth of the two eigenstates is simply given by $\gamma = (\gamma_{1S} + \gamma_{m,l})/2$. Where γ_{1S} and $\gamma_{m,l}$ are the QE and WGM linewidth. Even having taken into account decrease of the Q factor due to PMS contact with the cuprous oxide sample $\gamma_{1S} \gg \gamma_{1,39}$ for the ortho-exciton. Hence, γ is defined by the QE linewidth.

For the Cu_2O para-exciton the linewidth of the polariton is given by the WGM linewidth. The para-excitons can acquire some oscillator strength provided the PMS exerts a local stress upon the cuprous oxide sample. The applied stress changes the crystal symmetry, so that usually optically inactive para-excitons may couple to the WGM. Note that due to spin-orbit interaction the para-excitons are $12 \mu\text{eV}$ below the ortho-exciton. Therefore for given radius of the PMS the resonant interaction occurs with different WGM for ortho- and para-cases.

The evanescent polariton provided by a single sphere gives the time coherence necessary for the observable BEC of the quadrupole exciton. But the spatial coherence is limited to a small region near the sphere. To improve the spatial coherence one has to sacrifice the temporal coherence slightly by delocalizing the corresponding WGM. It can be done by using an array of spheres aligned along the z direction and separated by the distance δr_0 (See Fig.2).

Recent experimental [14] and theoretical [15] studies have shown that the WGM can travel along the chain as "heavy photons". Therefore the WGM acquires the spatial dispersion, and the evanescent quadrupole polariton has the form (See Fig.3):

$$2\omega = \omega_{1l,k} + \omega_{1S} \pm \sqrt{(\omega_{1l,k} - \omega_{1S})^2 + 4|g_{1l}/\hbar|^2} \\ \omega_{1l,k} = \omega_{1S} + 2\left(g_{1l}^l/\hbar\right)\cos(x - x_{1l} + \pi/2) \quad (9)$$



Figure 2: Schematic of formation of the evanescent polariton on linear chain of PMS. The actual dispersion is determined by the ratio of two coupling parameters such as exciton-WGM coupling and WGM-WGM coupling between the microspheres.

Here $g_{ll}^{ll} = \omega_{1S} b_{1l} A_{1l}^{ll} (\delta r_1)$ is the nearest-neighbor inter-sphere coupling parameter.

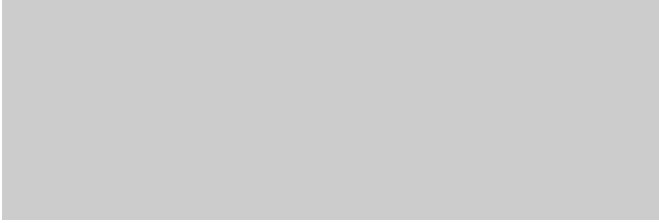


Figure 3: Dispersion of the evanescent polariton 39TE1. The dashed line (1) corresponds to the dispersion of the chain of spheres touching each other ($\delta r_0 = 0$). The thin solid line (3) stands for upper and lower branches of a single sphere dispersion ($\delta r_0 \gg \delta r = 0$). The thick solid curve (2) is the case of linear chain of the spheres in contact with the cuprous oxide ($\delta r_0 = \delta r = 0$).

When the coupling between spheres dominates ($\delta r \gg \delta r_0$) the minimum of the lower polariton branch disappears. Consequently, for possible BEC of the evanescent polariton one has to keep the desired balance between spatial and temporal coherence by adjusting experimental parameters δr and δr_0 .

Both, the energy of the $1S$ quadrupole exciton and the WGM depend on the temperature. Therefore one can use a standard temperature scan to reveal the evanescent polariton dispersion [16].

In summary, we note that there is some similarity between BEC of alkali atoms trapped by the laser field [17] and the long living QE localized by the resonant WGM.

The theory developed above is applicable also for void cavities, spherical impurities and metallic droplets in bulk cuprous oxide crystal.

4. Appendix

In the appendix we list explicit expression for the Mie scattering coefficient:

$$a_{ml} = \frac{n^2 j_{ml}(nx) [x j_{ml}(x)]' - j_{ml}(x) [nx j_{ml}(nx)]'}{n^2 j_{ml}(nx) [x h_{ml}^{(1)}(x)]' - h_{ml}^{(1)}(x) [nx j_{ml}(nx)]'}$$

$$b_{ml} = \frac{j_{ml}(nx) [x j_{ml}(x)]' - n^2 j_{ml}(x) [nx j_{ml}(nx)]'}{j_{ml}(nx) [x h_{ml}^{(1)}(x)]' - n^2 h_{ml}^{(1)}(x) [nx j_{ml}(nx)]'}$$

Here $n = \epsilon^2$ is the refractive index of the spheres; $x = kr_0$ is the size parameter; j_{ml} , h_{ml} are the spherical Bessel and Hankel of the first kind functions respectively.

In the case of $l \gg 1$ the calculation of the translational coefficients can be significantly simplified with the help of the so-called maximum term approximation [9].

$$A_l' \cong -2l(-1)^{l+1} \sqrt{\frac{l+l'}{\pi(l'+1)(l-1)}} \times$$

$$\frac{l'(l')^l}{(l'+1)^{l'+1}(l-1)^{l-1}} h_{l+l'}^{(1)}(\eta x)$$

$$B_l' \cong i \frac{x|i-j|}{ll'} A_l'$$

Here η defined as $\eta = |r_0 + \delta r|/r_0 \geq 1$ is a dimensionless distance between the centers of the spheres.

References

- [1] G. Kavoulakis and G. Baym, Phys. Rev. B **53**, 7227 (1996).
- [2] O. Roslyak and J. Birman, arXiv:cond-mat/0703650, PRB to be published (2007).
- [3] D. Frohlich, G. Dasbach, G. B. Hogerthal, M. Bayer, R. Kliebera, D. Sutura, and H. Stolz, Solid State Communications **134**, 139 (2005).
- [4] C. Ell, A. L. Ivanov, and H. Haug, Phys. Rev. B **57**, 9663 (1998).
- [5] D. Snoke, Science **298**, 1368 (2002).
- [6] J. Kasprzak, M. Richard, S. Kundermann, A. Baas, P. Jeambrun, J. Keeling, F. Marchetti, M. Szymanska, R. Andre, J. Staehli, et al., Nature **443**, 409 (2006).
- [7] H. W. Xudong Fan, Scott Lacey, Optics Letters **24**, 771 (1999).
- [8] X. Fan, S. Lacey, and H. Wang, Opt. Lett **24**, 771 (1999).
- [9] H. Miyazaki and Y. Jimba, Phys. Rev. B **62**, 7976 (2000).
- [10] C. Bohren and D. Huffman, *Absorption and scattering of light by small particles* (Wiley New York, 1983).
- [11] S. Stein, Q. appl. Math **19**, 15 (1961).
- [12] K. Fuller, Appl. Opt **30**, 4716 (1991).
- [13] H. J. Carmichael, Phys. Rev. A **33**, 3262 (1986).
- [14] Y. Hara, T. Mukaiyama, K. Takeda, and M. Kuwata-Gonokami, Physical Review Letters **94**, 203905 (2005).
- [15] L. Deych and A. Roslyak, Physical Review E **73**, 36606 (2006).
- [16] E. Peter, P. Senellart, D. Martrou, A. Lemaître, J. Hours, J. Gérard, and J. Bloch, Physical Review Letters **95**, 67401 (2005).
- [17] A. J. Leggett, Rev. Mod. Phys. **73**, 307 (2001).

Remark 3. We remark that; when the ratio $h = \frac{r-s}{4}$ tends to 0, the expression $L(r, s) = \frac{(s-r)^2}{(4h^2 + (r-s)^2)}$ tends to $1 = (r-s)$ which is a singular function. This means that the expression $L(r, s)$ is not well behaved for the small values of $h = \frac{r-s}{4}$. Consequently, for the solution to converge, the integrals of (10) and (11) must be evaluated with a large number of nodes. In our numerical applications (cf. section 5), we use 100 nodes to evaluate these integrals. With the smallest value of $h = \frac{r-s}{4} = 0.02$, the convergence is good with $N = 20$.

Theorem 2. For system (8), consensus can be achieved with $\|T_{1z}(s)\|_1 < \infty$ if there exist a symmetric positive definite matrix $P \in \mathbb{R}^{(n-1) \times (n-1)}$ and a scalar $\gamma > 0$ satisfying

$$= \begin{bmatrix} \bar{L}^T P & P\bar{L} + U_1^T U_1 + \bar{E} & P U_1^T E_1 \\ E_1^T U_1 P & I & 0 \\ U_1 P & 0 & I \end{bmatrix} < 0; \quad (10)$$

where $\bar{L} = U_1^T L U_1$ and $\bar{E} = U_1^T E_2^T E_2 U_1$:

Proof of Theorem 2. Proof follows straightforward from Lemma 3 and Theorem 1. However, it should be emphasized that all possible $L(t)$ should share a common Lyapunov function $V(\cdot) = \bar{x}^T(t)P(t)\bar{x}(t)$ (see the proof of Lemma 3 in Appendix A).

1. The enumerate environment starts with an optional argument '1.' so that the item counter will be suffixed by a period.
2. You can use '(a)' for alphabetical counter and '(i)' for roman counter.
 - a) Another level of list with alphabetical counter.
 - b) One more item before we start another.
 - (i) This item has roman numeral counter.
 - (ii) Another one before we close the third level.
 - c) Third item in second level.
3. All list items conclude with this step.

Step 1. This is the first step of the example list.

Step 2. Obviously this is the second step.

Step 3. The final step to wind up this example.


```
\includegraphics[width=3in,angle=45]{tiger.pdf}
```

Fig. 1. More details on the usage of `\includegraphics` can be found in the `grfguide.ps` of the $\text{\LaTeX}$ documentation.

[1] Knuth, D.E., *TeX: The Program*, Computers & Typesetting; B., 1995, Addison-Wesley Publishing Co., Inc., New York.